Shaping processes for $\underbrace{\text{polymers}}_{\text{similar to CASTING processes} \Rightarrow}$

Groover M.P.: Chapter 10

Plastic Products

to create many kind of possible objects

Plastics can be shaped into a wide variety of products:

very different objects created

- Molded parts
- Extruded sections
- Films
- Sheets
- Insulation coatings on electrical wires
- Fibers for textiles

physical components/blocks related to adesive ecc...

polymers to make part

In addition, plastics are often the principal ingredient in other materials, such as:

- Paints and varnishes
- Adhesives
- Various polymer matrix composites

Many plastic shaping processes can be adapted to produce items made of rubbers and polymer matrix composites.

Plastic Shaping Processes are Important

we speak about

|| POLYMER MELTS ||
bring materials to fluid state with high VISCOSITY STATE ...
behave like liquid

Almost unlimited variety of part geometries.

Plastic molding is a *net shape* process:

- Further shaping is not needed.

Less energy is required than for metals due to much lower processing temperatures:

- Handling of product is simplified during production because of lower temperatures.

Lower temperature respect to melting / recrystaliz
→ smaller heat cost

Painting or plating is usually not required.

↑ usually processes with no further processes (NET SHAPE PROCESS)

(COLORS → as additional chemicals (NOT necessary to paints later)
→ NOT need a final painting
color part of melt polymer

additional chemicals inside material shaped, different color of the object easy to have

POLYMER MELTS

To shape a thermoplastic polymer it must be heated so that it softens to the consistency of a liquid. from a THICK LIQUID

In this form, it is called a *polymer melt*. ≃ behav. like liquid

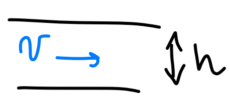
Important properties of polymer melts:

- Viscosity (necessary for complex geometry) → lower viscosity ↓ better for mold with thin/accurate cavity geometry
- Viscoelasticity ↑ particular phenomenon

Due to its high molecular weight, a polymer melt is a thick fluid with high viscosity.

Most polymer shaping processes involve flow through small channels or die openings. Flow rates are often large, leading to high shear rates and shear stresses, so significant pressures are required in these processes.

shear rate $\dot{\gamma} = \frac{v}{h}$ of a fluid between two plates



VISCOSITY, SHEAR RATE AND TEMPERATURE

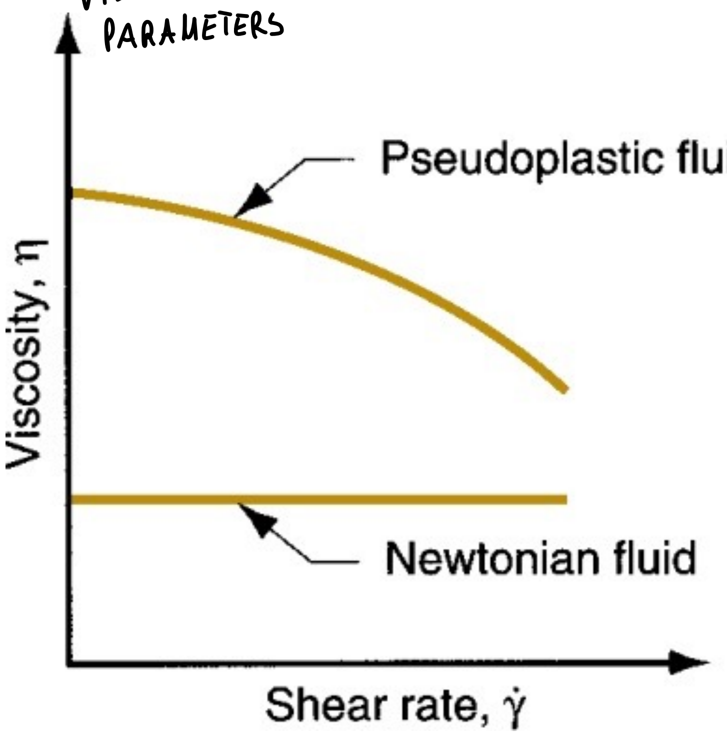
by increase T of the liquid, reducing viscosity η

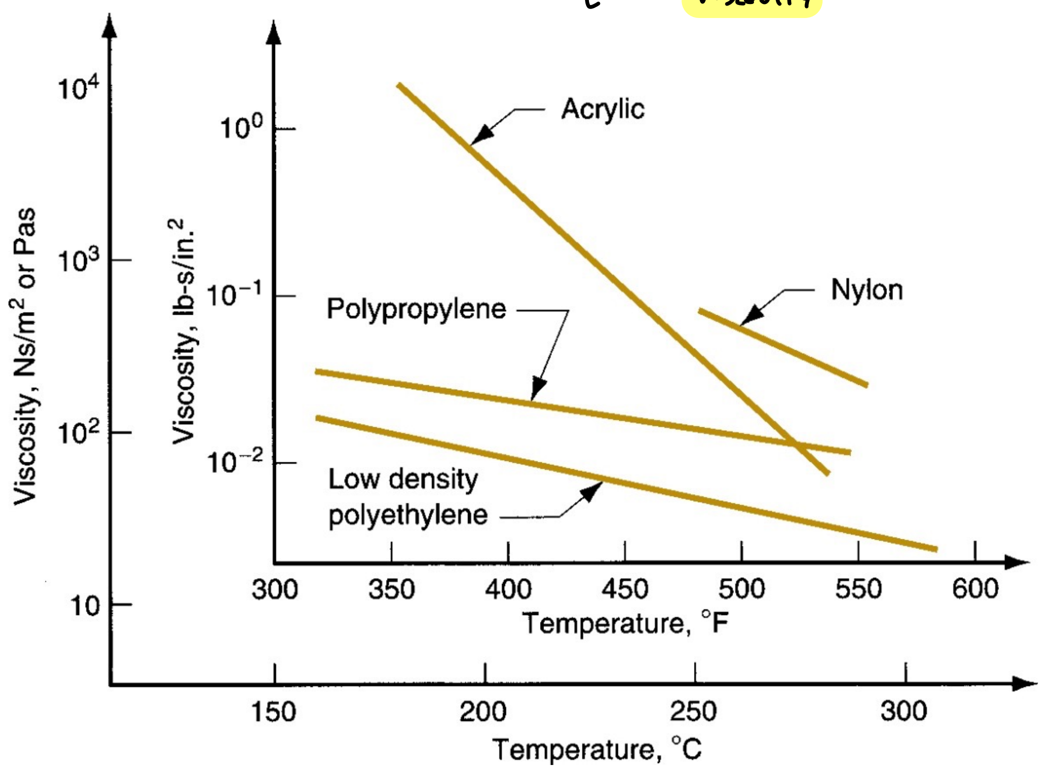
Viscosity of a polymer melt decreases with either shear rate or temperature. POLYMERS must create a sheere effect with viscosity consequences! (viscosity decrease irrespect of sheererate)

Thus the fluid becomes thinner (flows more easily) at either higher shear rates or higher temperatures.

if T↑↑ η↓↓ decrease viscosity

VISCOSITY PARAMETERS

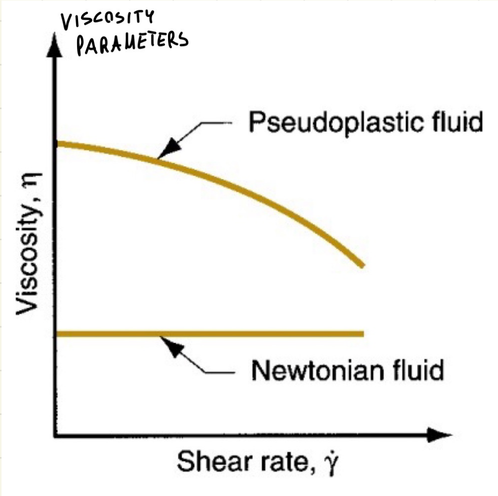




When shaping polymers → shearing STRESS due to flow inside DIES and channels

"shear Rate" has an effect · on VISCOSITY of polymer

polymer fluid → "PLASTIC" FLUID, viscosity Decrease respect shear rate $\dot{\gamma}$



increase T or $\dot{\gamma}$

to have a material with $\eta \downarrow$ VISCOSITY REDUCTION of polymer melt ≃ liquid

VISCOELASTICITY

polymer melt is a material with macromolecula creating thick fluid ⇒ when flow through DIE OPENING

KIND of ELASTIC RECOVERY → VISCOELASTICITY

Combination of viscosity and elasticity

Example: *die swell* in extrusion - hot plastic expands when exiting the die opening.

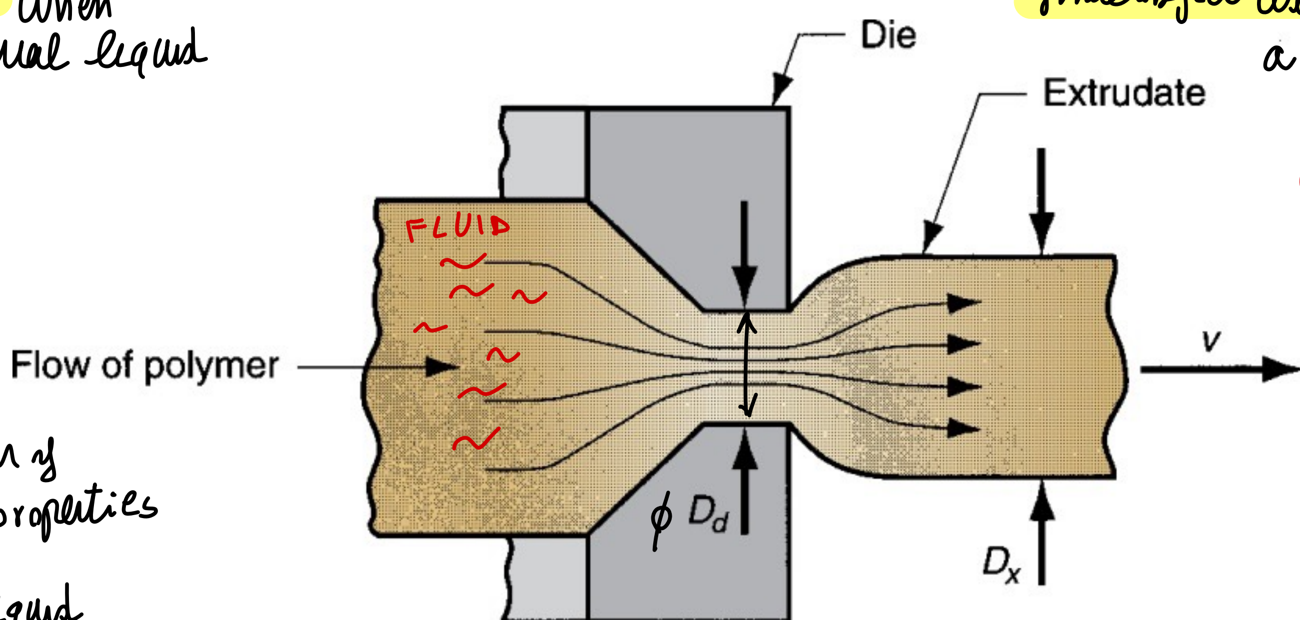
LIQUID Polymers

Goes through a Die as an open hole of Dd → final object will have a Dx>Dd

due to this VISCOELASTICITY

like an ELASTIC RECOVERY when the material liquid flows!

properties ↓ similar of SOLID properties even if liquid material polymer



VISCOSITY Play a relevant role

EXTRUSION

considering these BEHAVIOURS.. MAIN PROCESSES ARE..

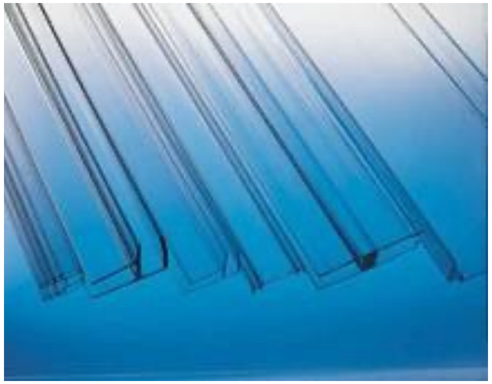
BASICALLY: defined the CROSS section, object with costant CROSS-SECT → cut in small pieces!

Material is forced to flow through a die orifice to provide long continuous product whose cross section is determined by the shape of the orifice.

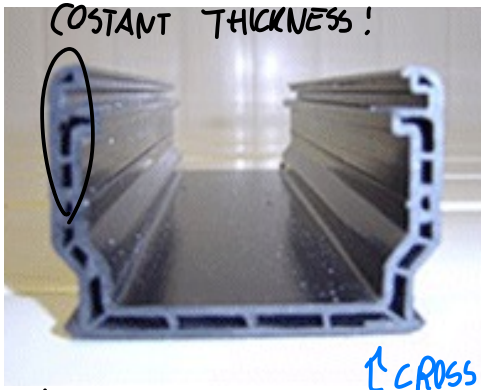
EXTRUSION: defined as BULK Metal forming process, basic definition of extrusion... once defined CROSS-SECTION by a DIE, the object of COSTANT CROSS-SECTION, possible infinite lenght, you shape CROSS-SECT, you cut the EXTRUDED piece

- Widely used for thermoplastics and elastomers to mass produce items such as tubing, pipes, hose, structural shapes, sheet and film, continuous filaments, and coated electrical wire.
- Carried out as a continuous process; *extrudate* is then cut into desired lengths.

Goal → COSTANT CROSS-SECTION and lenght NON defined



wall ≃ COSTANT THICKNESS!



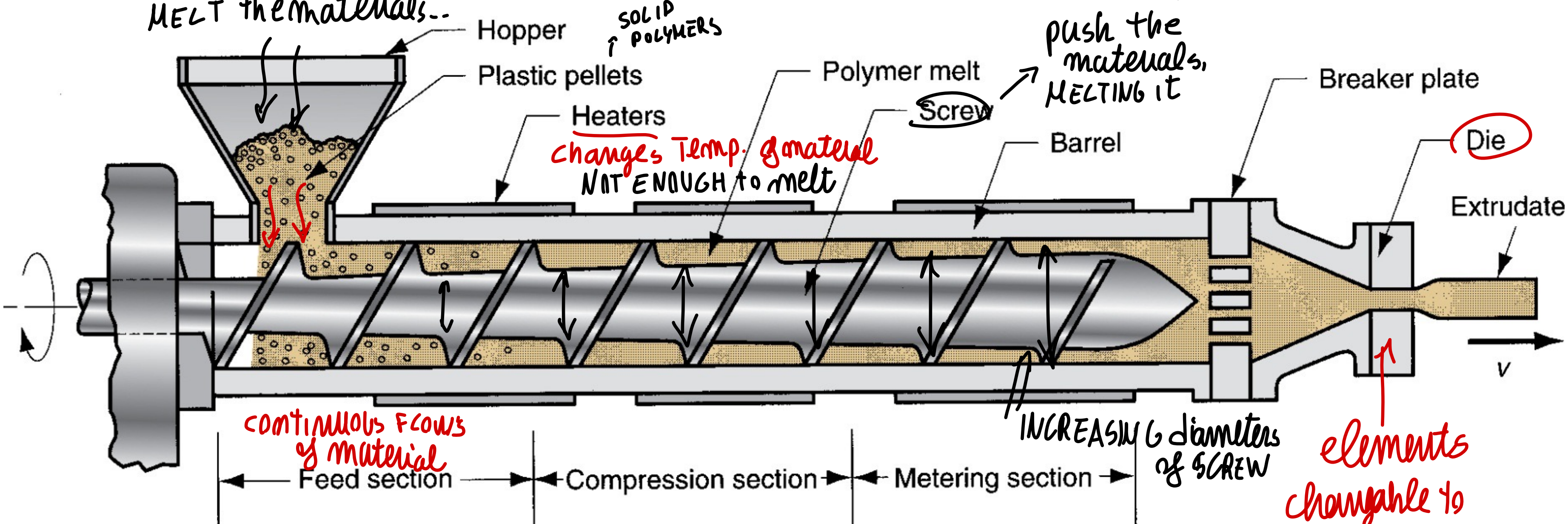
You can obtain COMPLEX Geometry by extrusion

↑ CROSS SECTION full of CAVITY

MACHINE TO DO EXTRUSION

EXTRUDER → machine which prepare the material

Components and features of a (single-screw) extruder for plastics and elastomers.
(SIMPLE CONFIGURATION)



Two main components of an extruder:

1. Barrel
2. Screw

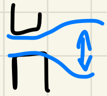
considering the VISCOELASTICITY, different shape from DIE!

The die is not an extruder component: It is a special tool that must be fabricated for the profile to be produced.

EXTRUDER

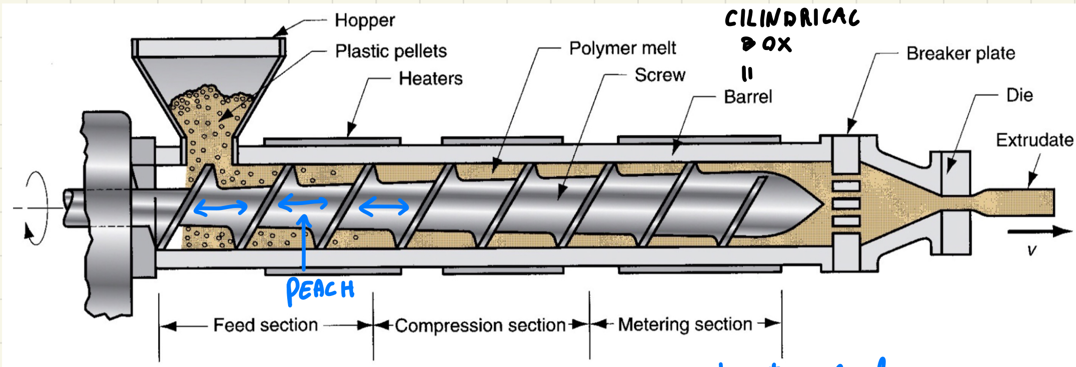
(theoretically ∞ lenght object)

DIE: changable element used to shape the object extruded that create CROSS-SECTION



VISCOELASTICITY
keep in mind that the shape changes when exit

machine that prepare the polymer melt, that continuously flow through the DIE



SIMPLER CONFIGURATION → single SCREW extruder = costant peach (distance between elements)

Barell containg the SCREW → take solid material in one side, bring to melt and push on other side

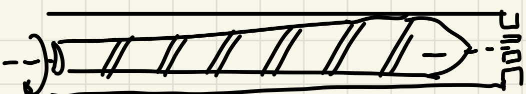
tipically
PELLETS of polymers
(thermoplastic ones) → bring inside the BARREL through an hopper
(solid polymer)

screw rotation move material inside

→ SOLID MATERIAL ⇒ POLYMER MELT thanks to

- HEATER on barell surface that increase T → melt help
- varing diameter of the SCREW

↳ increasing diameters towards the dies

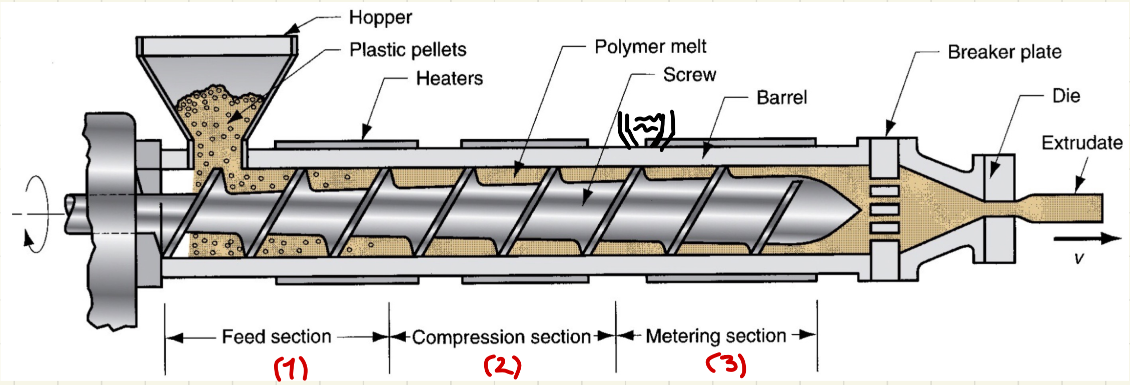


by INCREASING → melting solid polymer pellets

among we have air → lot of air space on volume

Volume between screw and barrel reduction towards dies ⇒ compress material

compressing material (less air volume)
↓
FRICTION due to COMPRESSION = Additional heating → full melting
compress → With VENTING HOLE : we remove GAS from material
(small holes where air flow OUTSIDE)
↑
all thanks to diameter increase of the SCREW



Barrel

can be DIVIDED on 3 section

1: when from Hopper we start to feed the Barrel

2: compressing material

3: omogeneize melted polymer → sometimes additional HOPPERS here, to add chemicals, ecc → MIX additional component

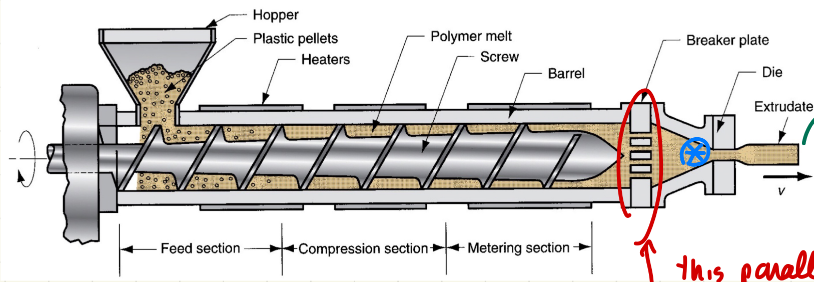
well compressed ⇓ omogeneous material ←

material pressure → to let it go through the DIES

BUT before DIE we have the SCREEN PACK

set of WIRES creating the NET, after it we have the BREAKER PLATE

With GOAL of FILTERING solid element remained, and INCREASE pressure!



COOL DOWN here to let material SOLIDIFY

this parallel holes with extruder direction STOP the ROTATION and let it move along

material keep rotating throw Barrel
↓
small holes on PLATE enable only linear motion

⊛ To have here enough PRESSURE to go through the DIE

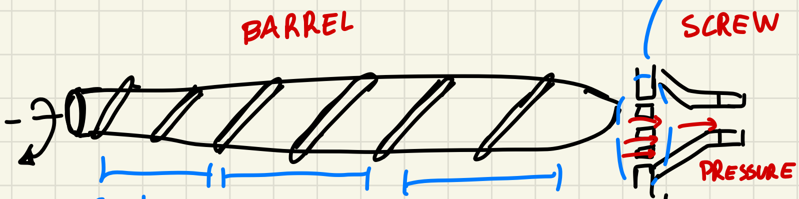
INCREASING
screw diameters so... every melting loosing the VOLUME...
we reduce the SPACE and COMPACT the MATERIAL ⇒ COMPRESSING the material

additional HEATING

With holes to let air go away

Reducing VOLUME to compress the POLYMER and get the AIR go OUTSIDE

STOP material ROTATION
let it enter directly linear!



feeding section
Where MATERIAL enter

COMPRESSION SECTION

METERING SECTION
omogeneus polymers to mix well the components

act as a filter, INCREASING the pressure

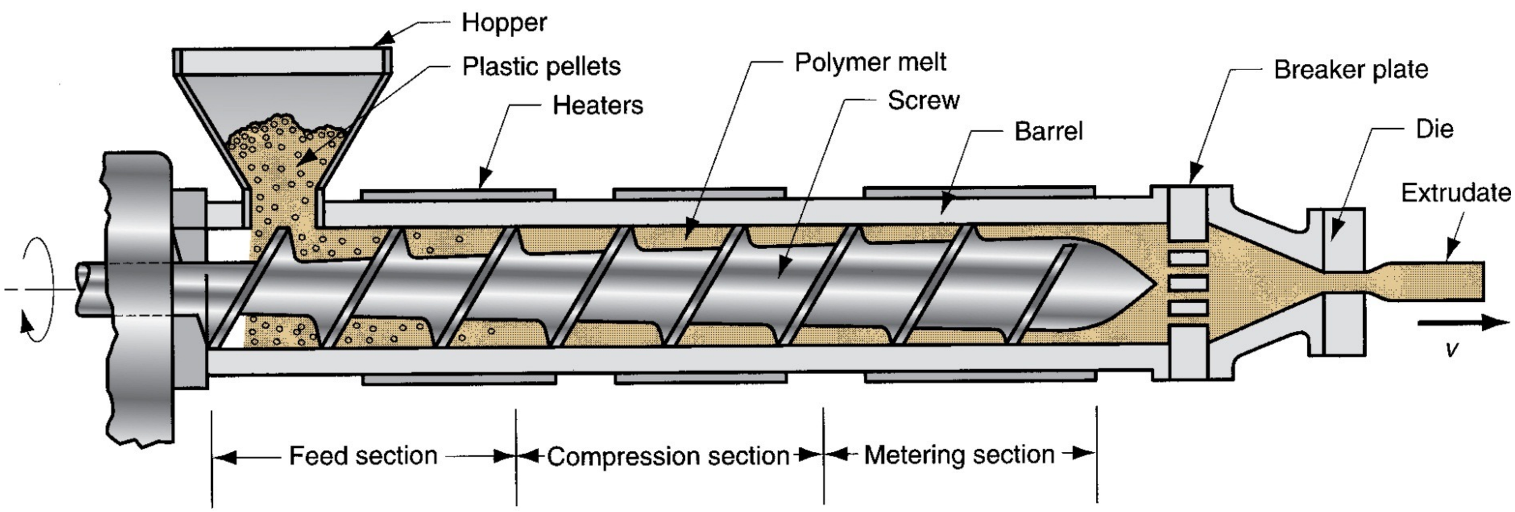
polymer melt stop to rotate thanks to BREAKER PLATE

enough pressure to die the object

after die → cool down to come back on SOLID STATE!

EXTRUDER BARREL (DETAILS...)

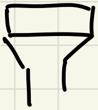
- Internal diameter typically ranges from 25 to 150 mm.
- *L*/*D* ratios usually ~ 10 to 30: higher ratios for thermoplastics, lower ratios for elastomers.
- Feedstock fed by gravity onto screw whose rotation moves material through barrel.
- Electric heaters melt feedstock; subsequent mixing and mechanical working adds heat which maintains the melt.



rotating screw brings the material inside

heating MELTING through HEATERS + venting holes to let air goes out

You can have secondary HOPPERS to let other materials come inside



According to final Die we can have different KIND of product

Recap:

Goal have an extruded (costant cross-section product)

let meterial through a DIE of COSTANT-SECTION

↑ with different geometries

- machining feedy by elastomers thermoplastic polymer pellets by an HOPPER

AIR INSIDE

- ROTATING screw bring material Solid INSIDE

- HEATERS + COMPRESSION (friction) (due to increase Diameter) ⇝ MELTING of polymer

+ VENTING HOLES on Barrel → AIR goes out

- METERING SECTION → omogeneous material

possible additive put inside by additional Hopper

↓

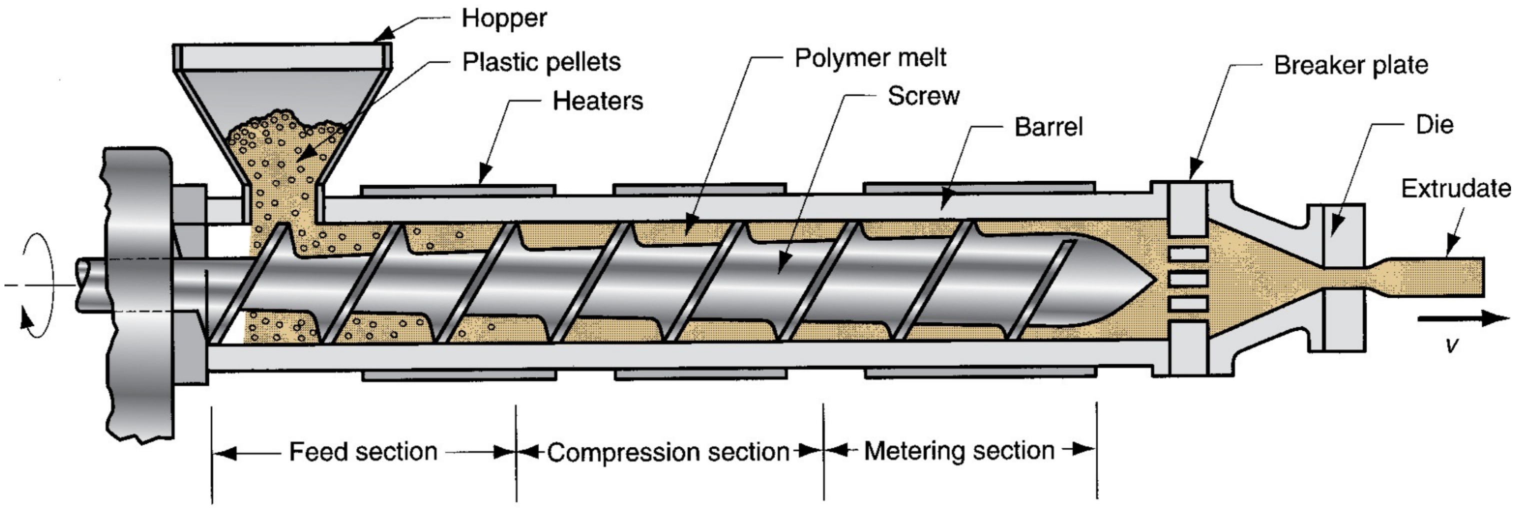
- SCREENPACK + BREAKER PLATE → remove impurities and make moovement linear

PRESSURE MATERIAL → through DIE → different product KIND

Extruder Screw

Divided into sections to serve several functions:

- Feed section - feedstock is moved from hopper and preheated.
- Compression section - polymer is transformed into thick fluid, air mixed with pellets is extracted from melt, and material is compressed.
- Metering section - melt is homogenized and sufficient pressure developed to pump it through die opening.

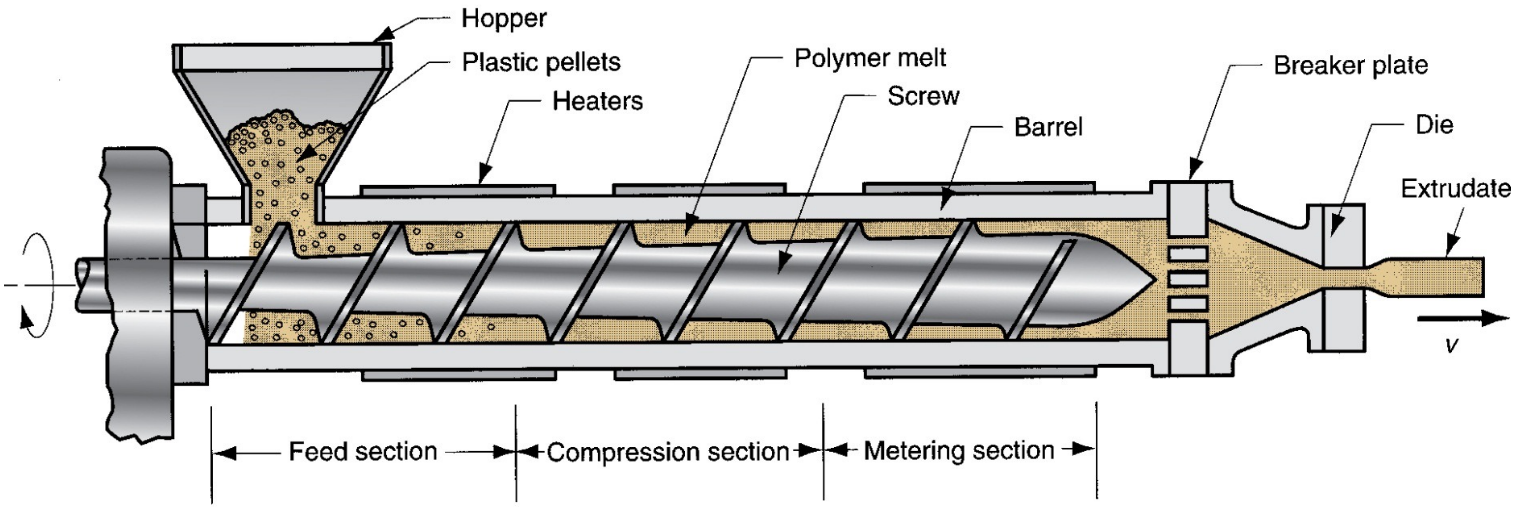


DIE END OF EXTRUDER

Progress of polymer melt through barrel leads ultimately to the die zone. Before reaching the die, the melt passes through a *screen pack* - series of wire meshes supported by a stiff plate containing small axial holes.

Functions of screen pack:

- Filters out contaminants and hard lumps
- Builds pressure in metering section
- Straightens flow of polymer melt and removes its "memory" of circular motion from screw



DIE CONFIGURATIONS AND EXTRUDED PRODUCTS

you can CREATE
VERY Differents Profile element

The shape of the die orifice determines the cross-sectional shape of the extrudate.

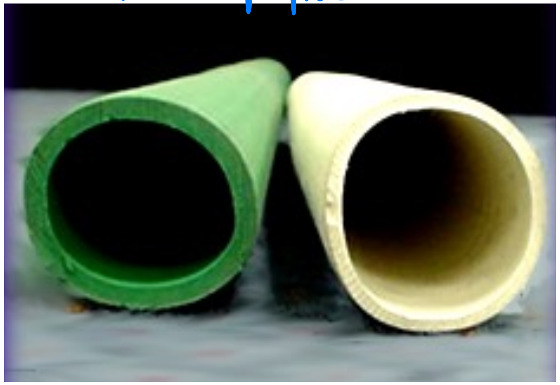
Common die profiles and corresponding extruded shapes:

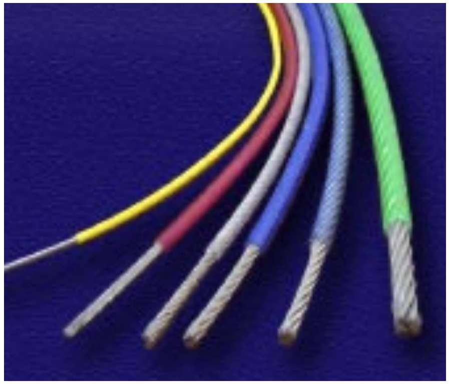
- Solid profiles
- Hollow profiles, such as tubes
- Wire and cable coating
- Sheet and film
- Filaments

from extruder, changing the final parts (DIE) I can obtain very different objects

different DIES to create different geometries obj

Hollow profile ("VUOTO")





EXTRUSION DIE FOR HOLLOW SHAPES

→ basic one is TUBOLAR shape

Side view of extrusion die for hollow cross sections.

polymeric tube with a given Di, De

Di De

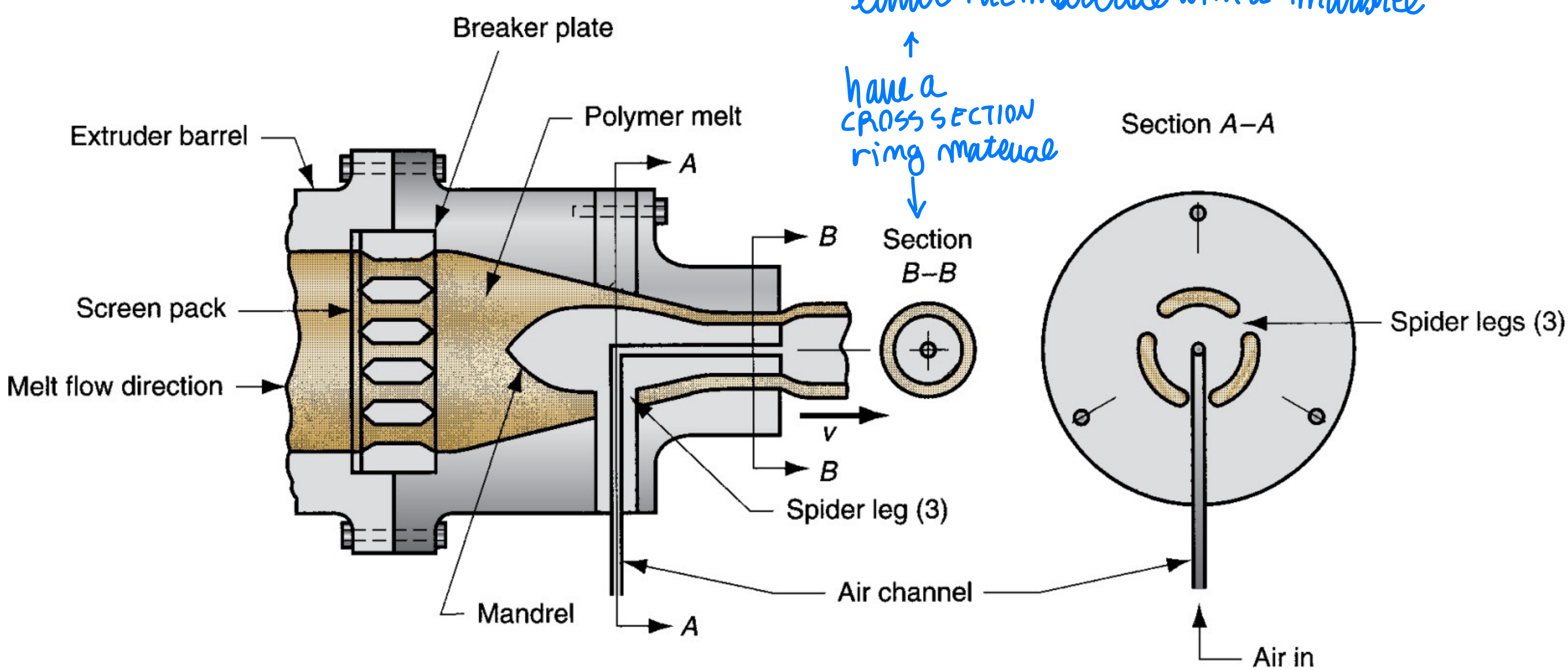
Section A-A is a front view of how mandrel is held in place.

Section B-B shows tube prior to exiting die; die swell enlarges diameter.

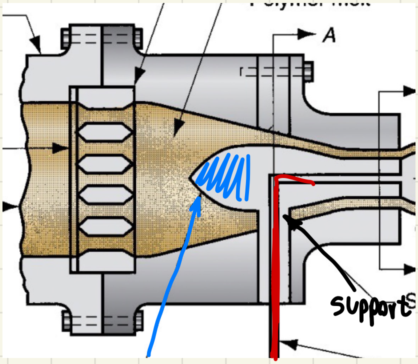
limit the material with a mandrel

↑

have a CROSS SECTION ring material



Goal: ring of MATERIAL as CROSS-SECTION

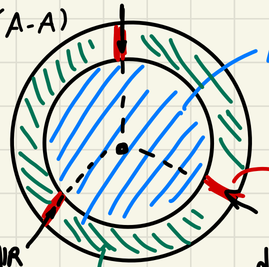


MANDREL to have a RING of material

↑ limit possibilities of flow material

you need a mandrel support

(A-A)



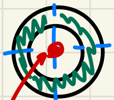
polymer melt has to be reshaped ← material passing

due to the link, material pass through the portion → both to support and bring compress air inside

on section A-A
we see 3 holes where material flow

Then an B-B cross section → solid element of mandrel yet supported

(B-B)



extrudate with tubular shape

IF material exit with shape B-B it COLLAPSE!

Inside the dies, there is a channel on mandrel where compressed air flow (AIR CHANNEL)

↓

pressure inside material to avoid collapse → stay in position to have solidification

section of mantryl with
3 holes section

here the material pass

tubolar shape of EXTRUDED material

BUT to avoid the collapse I need to change the mandrel

the channel inside mandrel, with compressed air avoid the materiae to collaps and stay in position

mandrel B-B section

AIR channels used to bring compressed air

With a complex mandrel shapes to create a Hollow shape of the material

EXTRUSION DIE FOR COATING WIRE

(ex: ELECTRICAL WIRE)
polymer COATING = "RIVESTIMENTO"

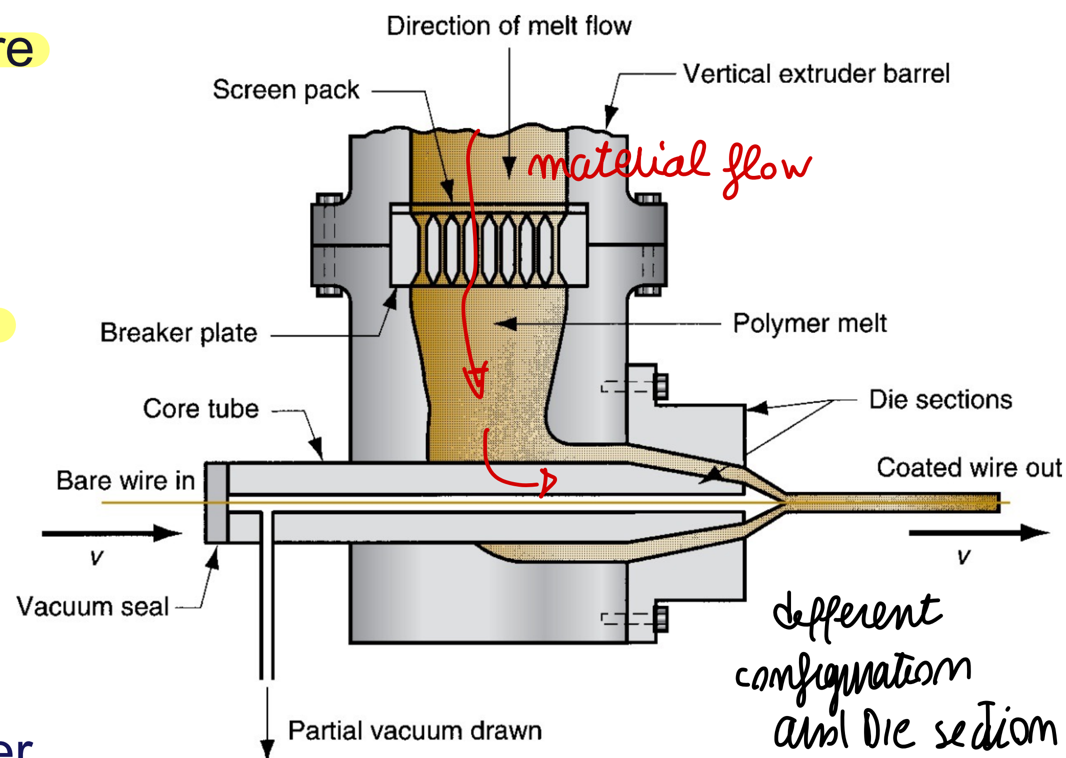
Polymer melt is applied to bare wire as it is pulled at high speed through die:

- A slight vacuum is drawn between wire and polymer to promote adhesion of coating.

Wire provides rigidity during cooling:

- Usually aided by passing coated wire through a water trough.

Product is wound onto large spools at speeds up to 50 m/s.



different configuration and Die section

to create a TUBULAR element around the wire and remove air later similarly removing air to create vacuum

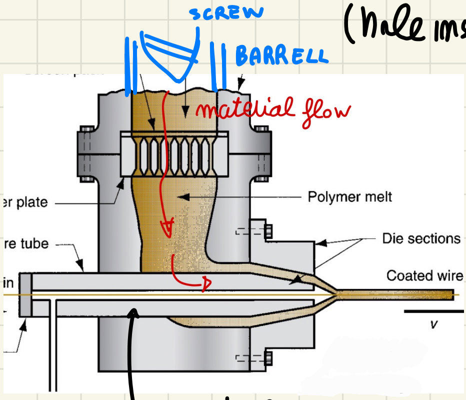
continuous wire feeding -> to let WIRE IN, extruder put in a vertical direction similar to Hollowshape → but here with $\underbrace{\text{WIRE INSIDE}}$

create tubular element around WIRE → let material collapse on the WIRE

↓

By **removing AIR!** **creating VACUUM**

(hole inside mandrel to remove AIR)



{wire feeding}

mandrel with CAVITY INSIDE

to have that mendrel

↓

different extruder direction → here **extruder IS VERTICAL**

↑

changing configuration and the DIE portion

Polymer Sheet and Film

DIE TO CREATE

define the element dimensions

Film - thickness below 0,5 mm

- Packaging - product wrapping material, grocery bags, and garbage bags
- Stock for photographic film
- Pool covers and irrigation ditch liners

shape of the die is important!

Sheet - thickness from 0,5 mm to about 12,5 mm

- Flat window glazing
- Thermoforming stock

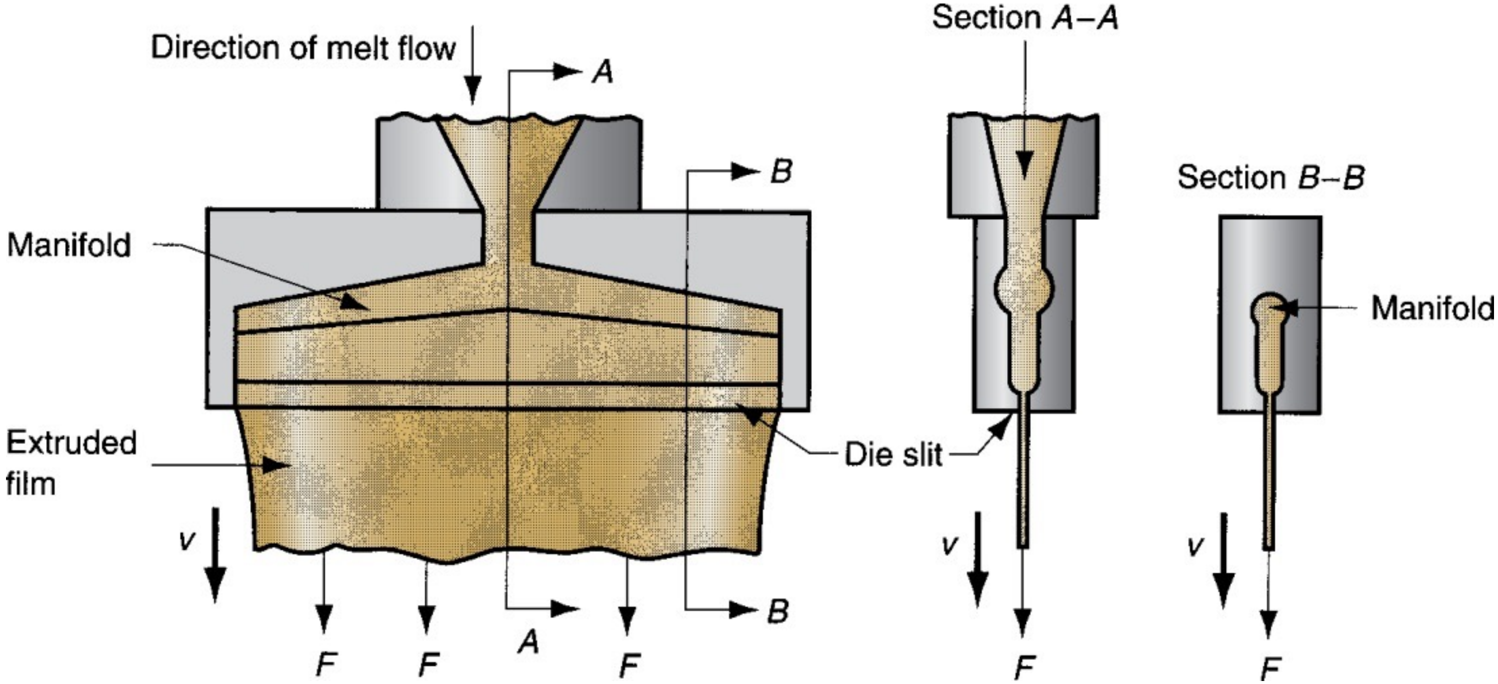
All thermoplastic polymers

- Polyethylene, mostly low density PE (LDPE)
- Polypropylene (PP)
- Polyvinylchloride (PVC)
- Cellophane

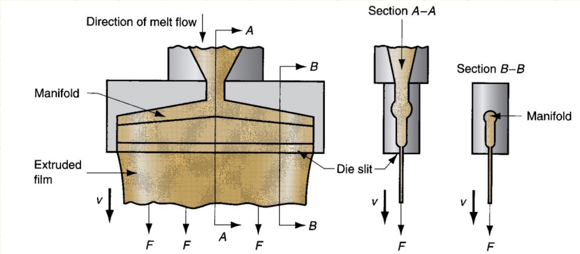
SLIT-DIE EXTRUSION OF SHEET AND FILM

Production of sheet and film by conventional extrusion, using a narrow slit as the die opening:

- Slit may be up to 3 m wide and as narrow as around 0,4 mm
- A problem is thickness uniformity throughout width of stock, due to drastic shape change of polymer melt as it flows through die
- Edges of film usually must be trimmed because of thickening at edges



DIE SHAPE



to CREATE a very thin layer of material

you can have small thickness but big slit ("fessura")

main extruder direction, like a down sprue

very thin CROSS-SECTION

omogeneusly bring material on all direction, while extruder on middle

FRONT VIEW

like to have down sprue and runners (similar metal-casting)

costant thick element,
very small SLIT → Die opening

long as considered part

CASTING

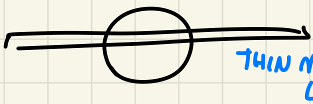
{ Runners → manifold
Gates → dies slit }

very thin layer

depending on thikness { sheet / Film }

to CONTROL omogeneity of thickness
(Difficult on edges → will be treated) + Elastic behav (different size than opening dimension)

To obtain very thin cross-section
⇓
bring material in all direction



distribute the material so that channel reducing the dimension

→ depends on thin SHEET or FILM

VERY small slit (die opening)

↓

different configuration

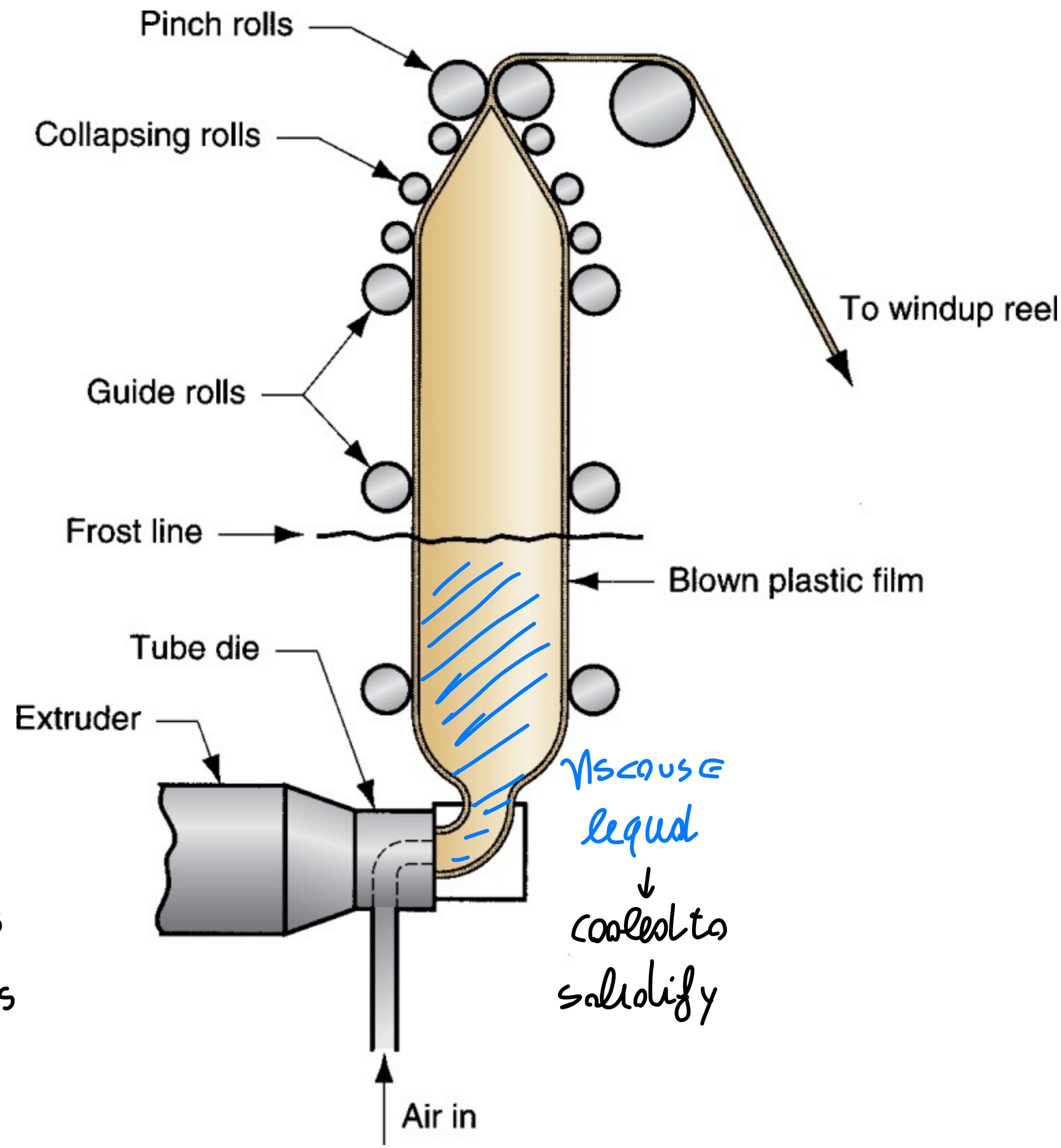
→ to controle omogeneity of thickness, edges are cut with different elastic behaviour

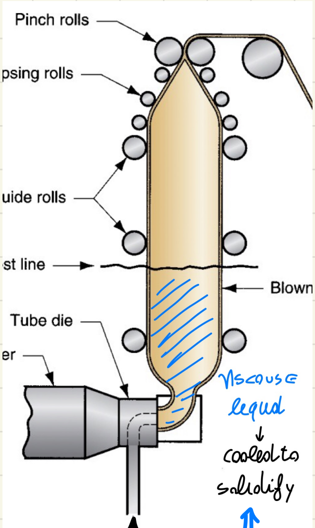
BLOWN-FILM EXTRUSION PROCESS

Combines extrusion and blowing to produce a tube of thin film

- Process sequence:
    1. Extrusion of tube
    2. Tube is drawn upward while still molten and expanded by air inflated into it through die
- Air is blown into tube to maintain uniform film thickness and tube diameter

Blowing the material, costant thikness of material, usually VERTICAL process





classical PROCESS to create PLASTIC (BELT ?)
(BASE ?)

→ tubular film that
we flaten and welt trasvertially

tubular element : die shape similar to seen

BUT then let AIR IN (lot of AIR)
to create a thin material layer
↳ BLOWING material

obtain costant thikness → (on vertical CROSS-SECT)

process on vertical direction

↑
material is cooled to solidify here...

→ but on that solidified material portion, we close the tubular element thanks to a rig of solid material

Usually PLANAR element to close the tubular element → Plastic base ⇐

CALENDERING

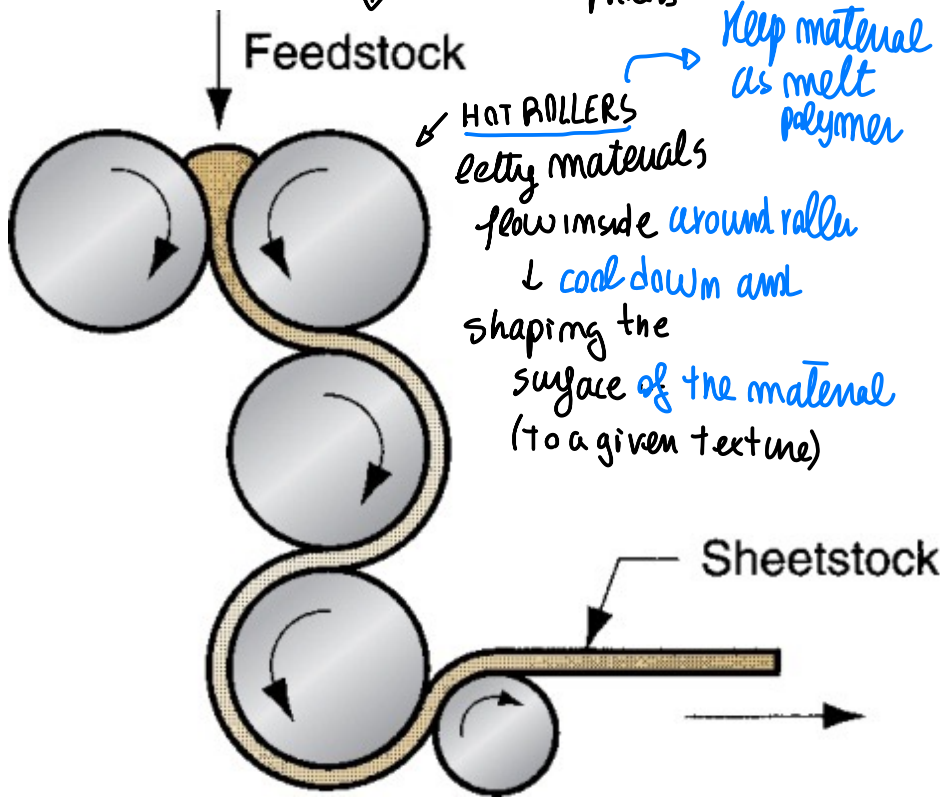
FEEDSTOCK comes from extrusion

Feedstock is passed through a series of rolls to reduce thickness to desired gage:

- Expensive equipment, high production rates.
- Process is noted for good surface finish and high gage accuracy.
- Typical materials: rubber or rubbery thermoplastics such as plasticized PVC.
- Products: PVC floor covering, shower curtains, vinyl tablecloths, pool liners, and inflatable toys.

With particular Texture usually

coming from extrusion process



Keep material as melt polymer

HOT ROLLERS

Jelly materials flow inside around roller

cool down and shaping the surface of the material (to a given texture)

@ The end → SOLID polymeric sheet of material

to create FLOOR material, particular texture other kind of output

Fiber and Filament Products

Definitions:

- *Fiber* - a long, thin strand whose length is at least 100 times its cross-section
- *Filament* - a fiber of continuous length

(continuous fiber)

small wire of material
use as small "building elements" inside materials (for textile, or reinforcement element inside composite material)

Applications:

- Fibers and filaments for textiles
- Reinforcing materials in polymer composites

Fibers can be natural or synthetic:

- Natural fibers constitute ~ 25% of total market
  - Cotton is by far the most important staple
  - Wool production is much less than cotton
- Synthetic fibers constitute ~ 75% of total fiber market
  - Polyester is the most important
  - Others: nylon, acrylics, and rayon

Comparison between NATURAL / SYNTETIC

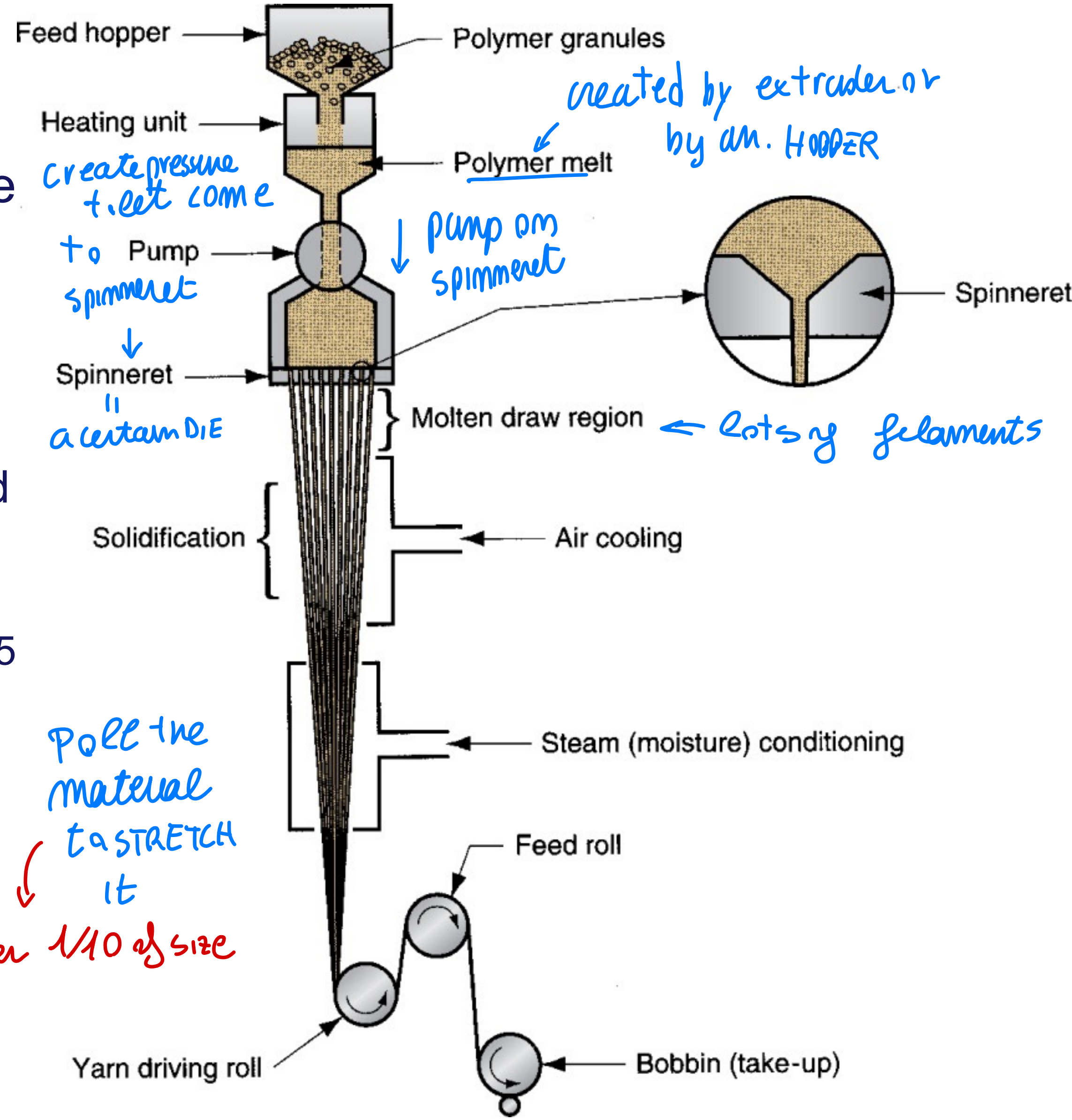
Fiber and Filament Production - Spinning

For synthetic fibers, *spinning* = extrusion of polymer melt or solution through a *spinneret* (die with multiple small holes), then drawing and winding onto a *bobbin*.

Melt Spinning: Starting polymer is heated to molten state and pumped through spinneret.

- Typical spinneret is 6 mm thick and contains ~ 50 holes of diameter 0,25 mm. (right pressure)
- Filaments are drawn and air cooled before being spooled onto bobbin.
- Final diameter wound onto bobbin may be only 1/10 of extruded size.
- Used for polyester and nylon filaments.

After extrusione → STILL   tensile test



created by extruder or by an. HOPPER

create pressure t. let come to spinneret

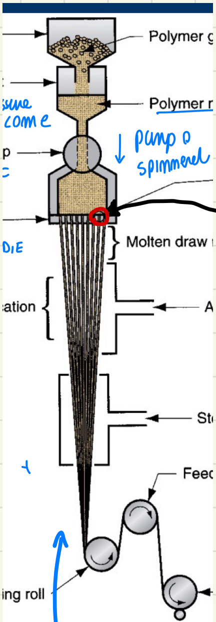
pump on spinneret

= a certain DIE

lots of filaments

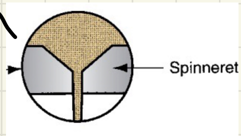
Pull the material to STRETCH it

fiber 1/10 of size



still deformable material

we melt polymer by an extruder OR
HERE after the HOPPER, in a Barrel we heat the polymer and bring to mealt
↓
pumped into the SPINNERET
≡
it is like a DIE with lot of small holes



PUMP create pressure to let it flow

more shaping process on that prolution

thick viscouse liquid → SOLIDIFICATION
⇒ FLAMENTS
↓
pulled by ROLLERS → STRETCH each filament

final Diameter of the fiber is
up to 1/10 of extruder size (holes of spimmeret)

usually 6 mm spimmeret and around 50 holes of ≈0.25 mm

RIGHT PRESSURE to let material go through thin holes

after die extrusion → TENSILE STRESS on the material
stretching further the element

and then WOUND material

... are extrusion processes defined by
the DIES shape

ANOTHER KIND OF PROCESS...

INJECTION MOLDING

one of MOST Important POLYMER SHAPING process (similar to CASTING) ≃ PERMANENT MOLD

bring polymer melt inside → open Mold when full solidification happen → extract the product and START AGAIN

Polymer is heated to a highly plastic state and forced to flow under high pressure into a mold cavity where it solidifies, and the *molding* is then removed from the cavity.

- Produces discrete components to net shape.
- Typical cycle time ~10 to 30 s, but cycles of one minute or more are not uncommon.
- Mold may contain multiple cavities, so multiple moldings are produced each cycle.

similar to CASTING on the process)

Injection molding is the most widely used molding process for thermoplastics.

Some thermosets and elastomers are injection molded:

- Modifications in equipment and operating parameters must be made to avoid premature cross-linking of these materials before injection.

Injection Molded Parts

Complex and intricate shapes are possible.

Shape limitations:

- Capability to fabricate a mold whose cavity is the same geometry as part;
- Shape must allow for part removal from mold.

Part size from ~ 50 g up to ~ 25 kg.

Injection molding is economical only for large production quantities due to high cost of mold.

INJECTION MOLDING MACHINE

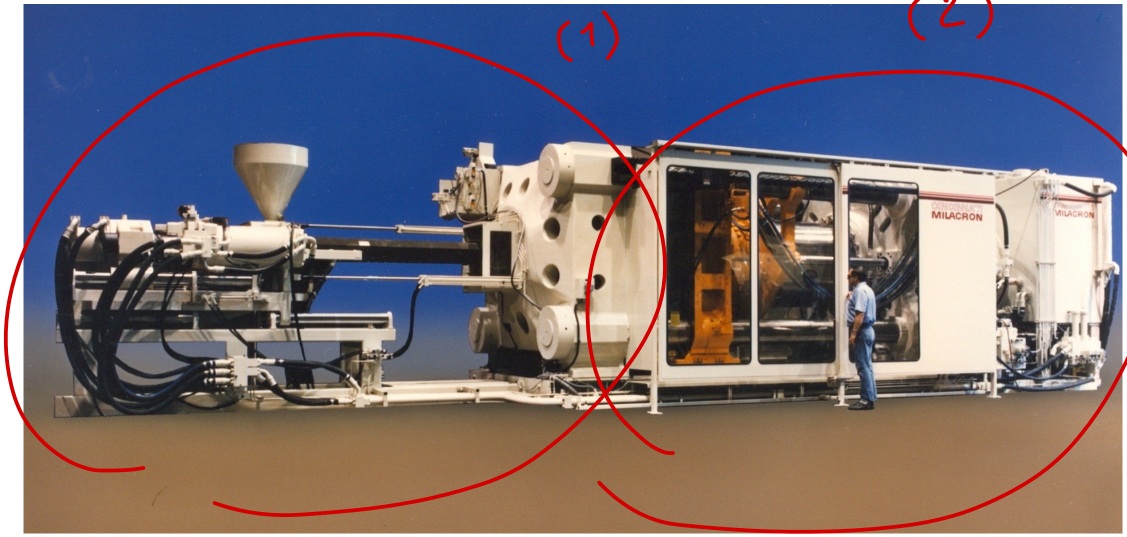
machine made of
2 BASIC ELEMENT

Two principal components:

1. Injection unit → prepare the material and responsable of injecting material inside mold
   - Melts and delivers polymer melt
   - Operates much like an extruder
2. Clamping unit (MOLD → issue of open/close the mold)
   - Opens and closes mold each injection cycle

have to create the right force to resist injection force creating by injecting material

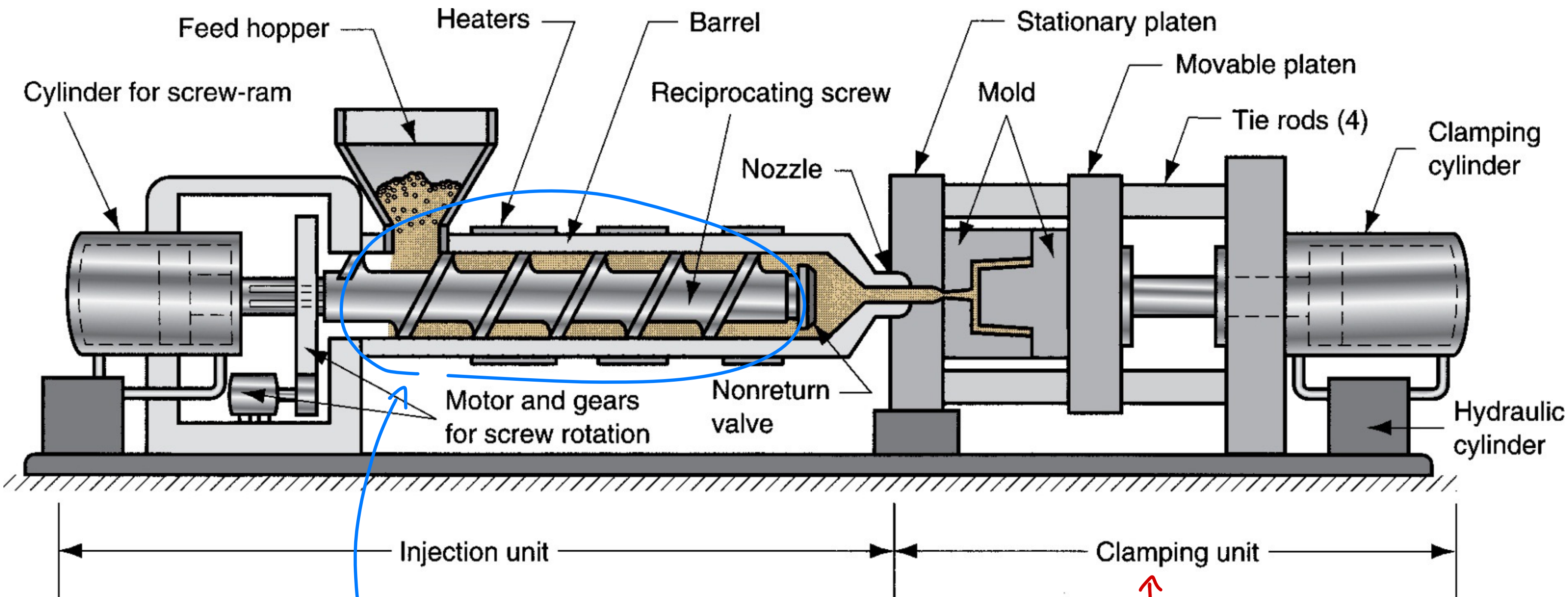
(1) (2)



Several clamping designs
- Mechanical;
- Hydraulic.

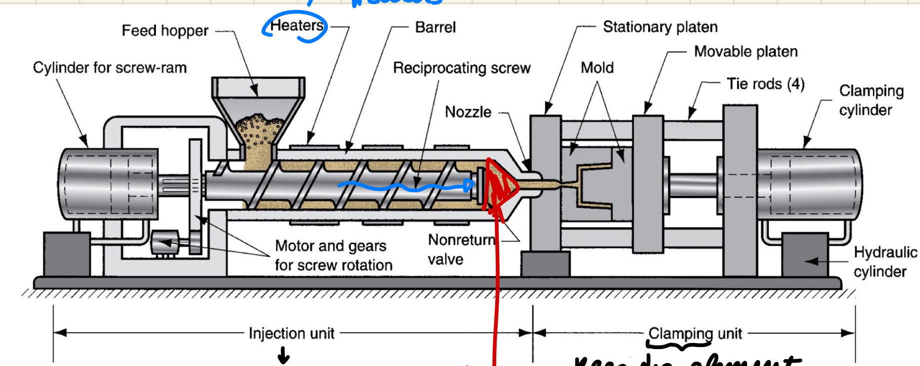
INJECTION MOLDING MACHINE

] schematic view...



similar to EXTRUDER but with a discontinuous flow of material

keeps die in position to resist the pressure electrical heaters



(*)

Injection unit ↓ prepare and INJECT material inside: similar to EXTRUDER

NOT CONTINUOUS flow of material respect extruder
DISCONTINUOUS process

Clamping unit: keep the element in position, resist injection force → GOAL open/close the mold ∀cycle

Idraulic actuator to have linear motion of the movable mold portion

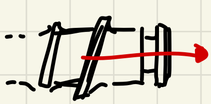
- PREPARING → to PREPARE the material similar to extruder... barrel, screw (constant pitch, variable diameters increasing)
- INJECTING

(+ compression remove air)

enough material here to fill the mold ← (*) big space in front of screw where material stop before the NOZZLE hole for injection

→ once MOLD is empty and closed INJECT material by the SCREW

INJECTION: screw stop rotating: moved along linear axis



NON RETURN VALVE to inject material inside mold cavity
full screw linear motion

the MOTOR allows ROTATION + LINEAR motion

once material start to solidify → enough solid to reduce mold pressure
↓
parallel process... { move back the SCREW in original position and start again material preparation during full solidification of polymer
↓
once open the mold we have enough material for new molding → max TROUGHPUT

INJECTION UNIT OF MOLDING MACHINE

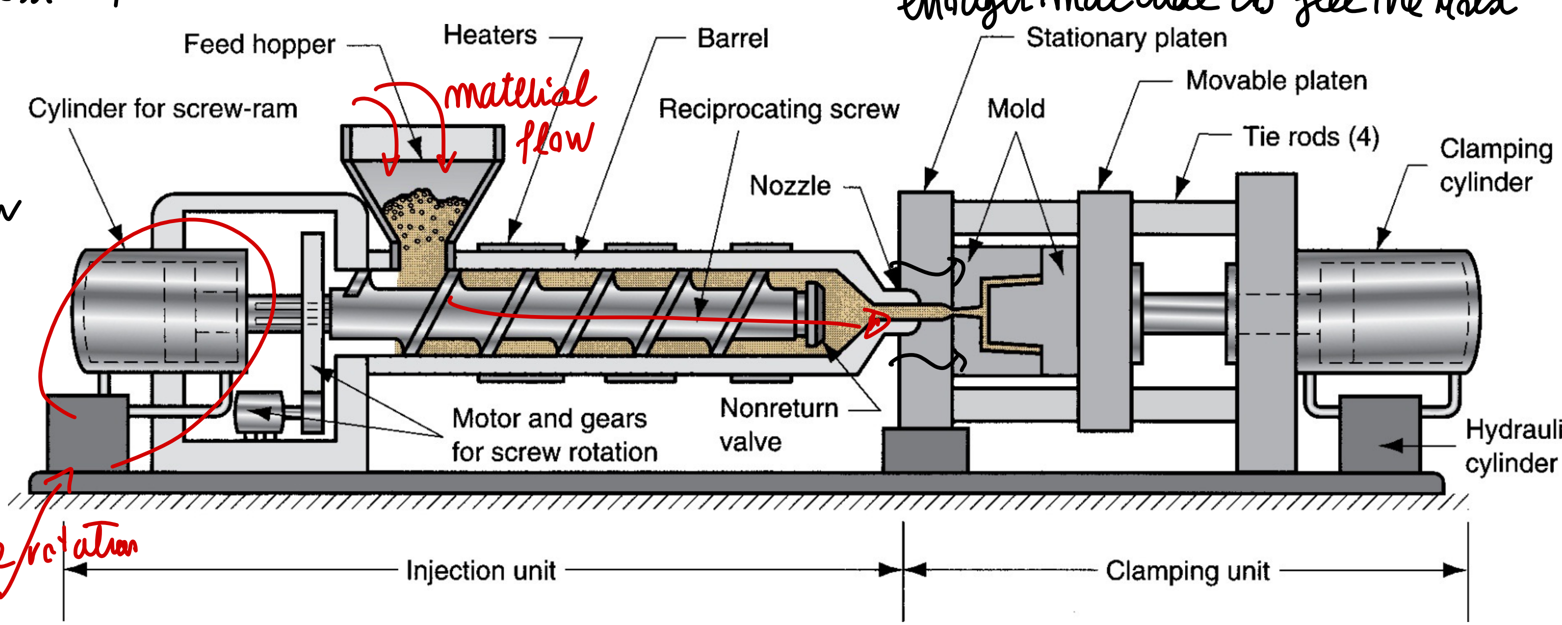
to prepare the material we have similar elements to EXTRUDER

Consists of *barrel* fed from one end by hopper containing supply of plastic pellets. Inside the barrel is a *screw* with two functions:

1. Rotates for mixing and heating polymer
2. Acts as a ram (i.e., plunger) to inject molten plastic into mold
   - Non-return valve near tip of screw prevents melt from flowing backward along screw threads
   - Later in cycle ram retracts to its former position

(remove air by compression)

enough material to fill the mold

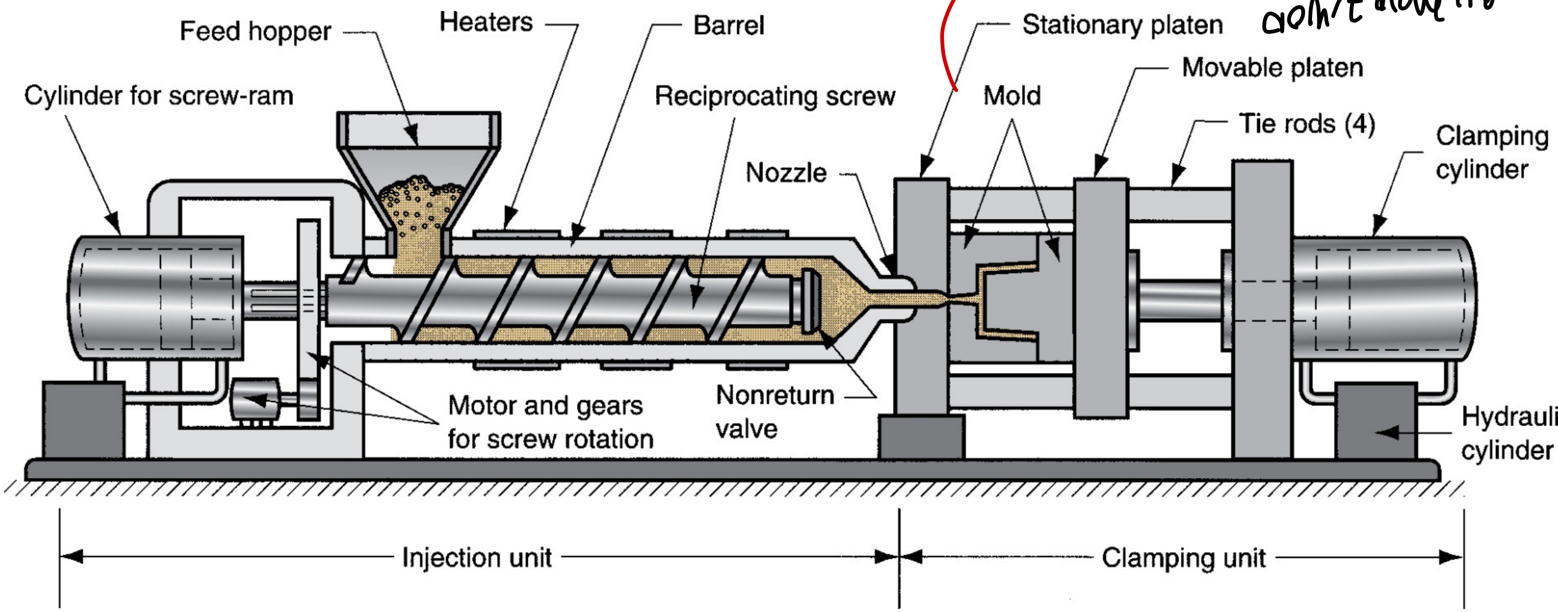


Clamping Unit of Molding Machine

Functions:

1. Holds two halves of mold in proper alignment with each other;
2. Keeps mold closed during injection by applying a clamping force sufficient to resist injection force;
3. Opens and closes mold at the appropriate times in molding cycle.

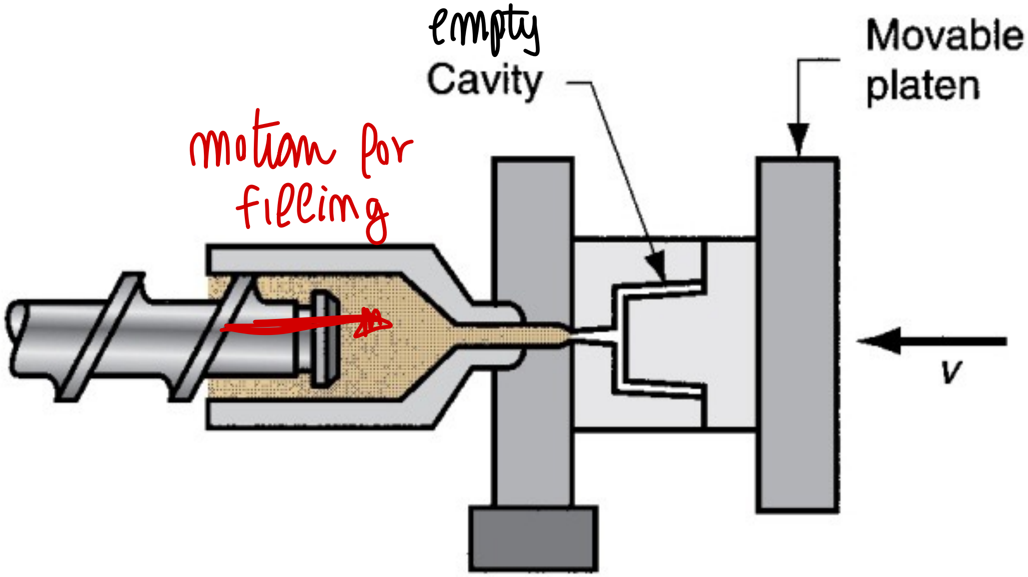
as soon as open the mold, we don't have more material



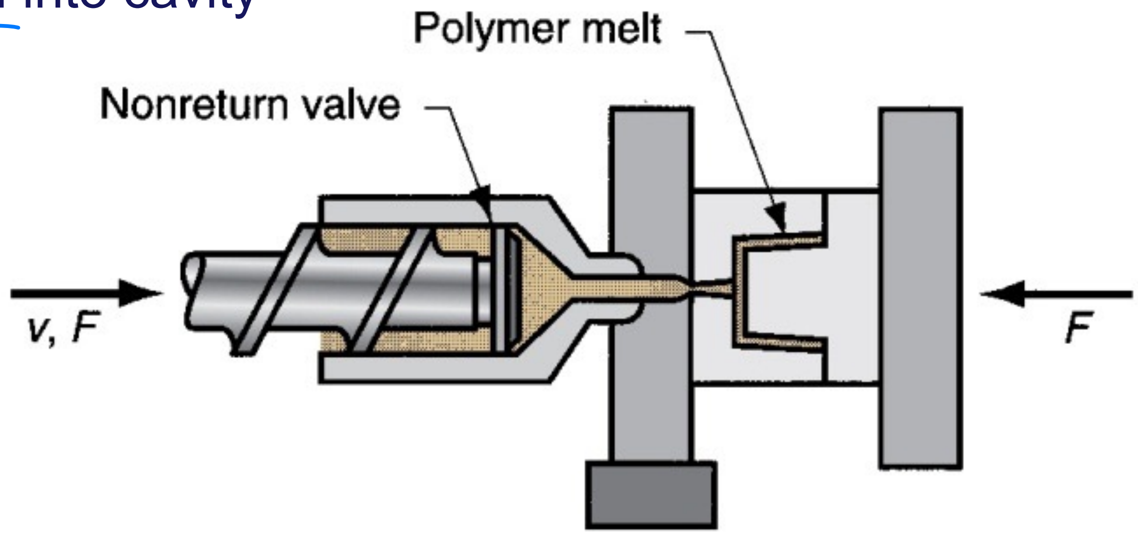
Injection Molding Cycle

Maximize THROUGHPUT

(1) Mold is closed



(2) Melt is injected into cavity

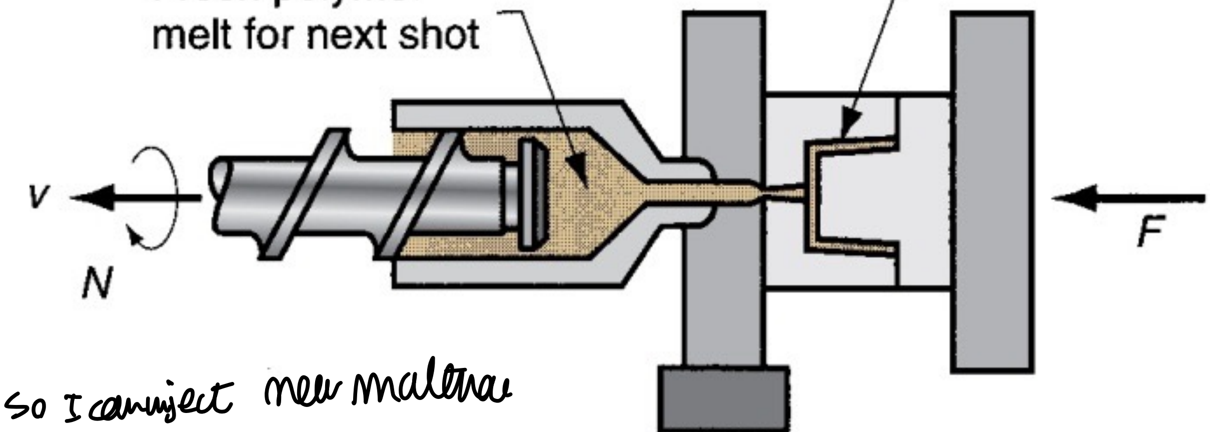


INJECTION MOLDING CYCLE

NB. On thermoset material can't be melted..
Inject polymer inside the mold

PROCESS time controlled by material used

3) screw is retracted

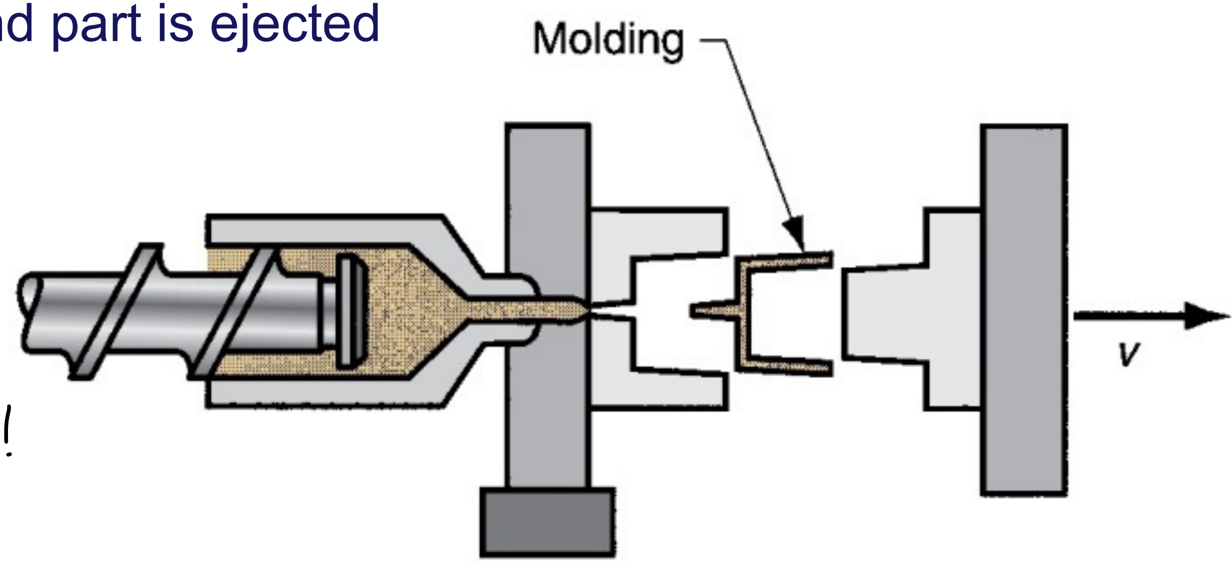


fasten the cooling than let this cool in air so I can inject new material
=> MAX the THROUGHPUT

(4) mold is opened, and part is ejected

extraction of full cooling parts!

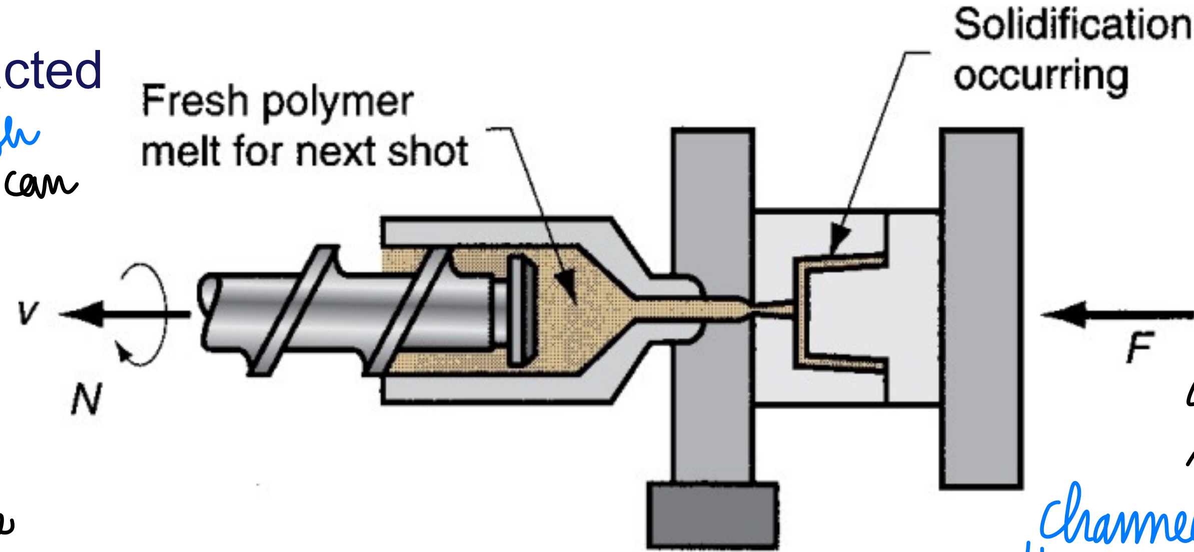
IMPORTANT to minimize time (the time to have the obj ready) → remove heat!
CONTROLLING of cooling process



INJECTION MOLDING CYCLE

3) screw is retracted

when solid enough POLYMER... screw can bring bag and start rotation



CONTROL COOLING

channels inside the mold with water to speed cooling through channels

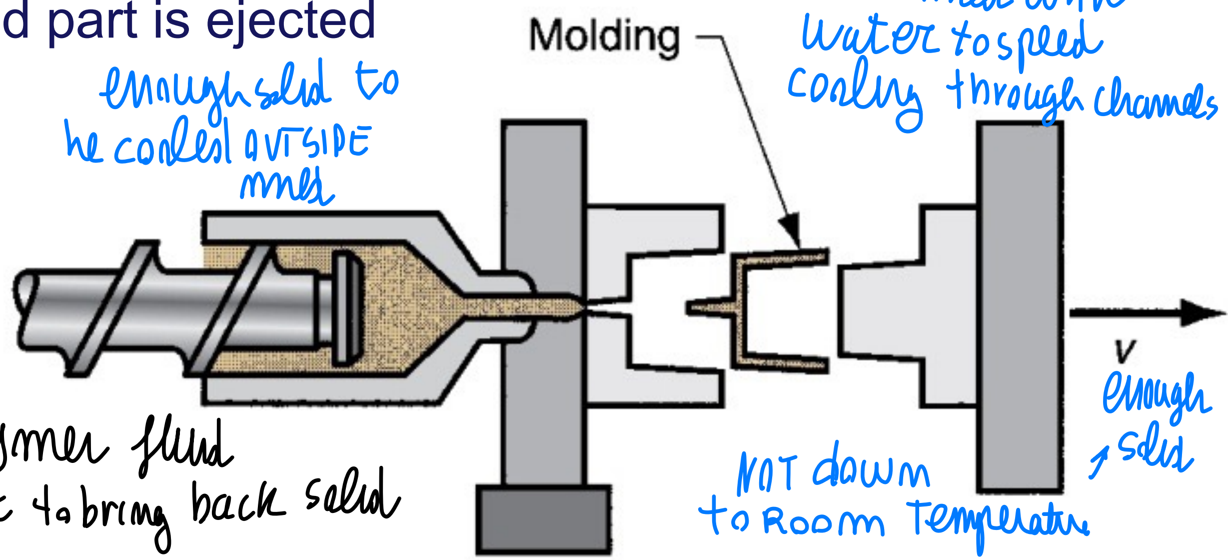
when close again, ready to pour material

(4) mold is opened, and part is ejected

extraction of part after FOOL COOLING

↳ ready to START (1)

enough solid to be cooled OUTSIDE mold



enough solid

NOT down to Room Temperature

minimize time is important to have enough solid object to be solid outside ⇒ thick polymer fluid

TERMOPLASTIC material: remove Heat to bring back solid

cooling control { faster solidification min{T_{TS}} total solidif.
bigger throughput max{THROUGHPUT}

possible for thermoplastic polymers in which we control { heating → melt
cooling → solidify

While on thermoset material → INJECTION MOLD?
yes, BUT CAREFULL..

polymer melt: pre-polymers liquid → Injected in MOLD
(thermoset cannot be melted)

giving energy to start polymerization....

you need to HEAT THE MOLD → COURING (chemical reaction)
depend on material and size

process time controlled by the material, NO CONTROL on total time ← you have to Weight chemical reaction

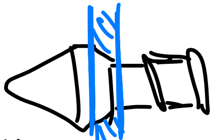
Full solidification → remain solid..

channel of MOLD to heat it and give energy to start polymerization

INJECTION MOLDING

LOW COST OBJECT but complex system → only for HIGH VOLUME PRODUCTION
LARGE-HIGH PRODUCTION VOLUME

filling the system through the hopper → enter the barrel and the screw flow it as melted



NON RETURN VALVE

don't allow to come back material

You can have very complex mold

relevant device of machine

open mold and extract part...

free material and let it go OUTSIDE
↓
double motion to create object
additional motion to let material OUT

↳ you can have a MULTI-CAVITY mold → channels ∀ cavity (multiple objects created at once)

one main injection NOZZLE and then multiple gates to separate material ∀ cavity independently

to remove solid objects → eject the piece thanks to a MOTION → EJECT part molded by rotating the internal part → additional rotation ∀ obj

SHRINKAGE

Accurate OUTPUT Dimension required
respect to casting SHRINKAGE here...

Polymers have high thermal expansion coefficients, so significant shrinkage occurs during solidification and cooling in mold.

to product
{NET-SHAPED object}
we need accurate OUTPUT dimensions

Plastic	*Typical Shrinkage, mm/mm*
Nylon-6,6	0,020
Polyethylene	0,025
Polystyrene	0,004
PVC	0,005

Dimensions of mold cavity must be larger than specified part dimensions:

CASTING

$$D_c = D_p + D_pS + D_pS^2$$

FORMULA

better control of product shapes to PRODUCE, we have an additional element

$(S D_p) S$: "SHRINKAGE of another SHRINKAGE

Where:

- D_c = dimension of cavity;
- D_p = molded part dimension;
- S = shrinkage value. ← we will have SHRINKAGE → here...

Third term on right hand side corrects for shrinkage in the shrinkage

$$D_c = D_p + D_p S + D_p S^2$$
CAVITY

comparing SOLID STATE shrinkage of CASTING

{ S = shrinkage allowance }

↓

to choose cavity dimension required.. D_c

$= D_p$ mold dimension

$+ D_p S$ related to shrinkage (measuring dimension) on 1 DIMENSION

------ in metal cast $D_p + S \cdot D_p$ is enough → NEAR NET process

to have good shape control... IN INJECTION MOLDING

We consider also $+ D_p S^2$

shrinkage of the shrinkage...

$(D_p S) \cdot S \Leftarrow S^2 D_p$

to have better final shape CONTROL

Shrinkage Factors

controlled by the packed Material inside the mold quantity

- Fillers in the plastic tend to reduce shrinkage.
- Injection pressure – higher pressures force more material into mold cavity to reduce shrinkage.
- Compaction time – similar effect – longer time forces more material into cavity to reduce shrinkage.
- Molding temperature - higher temperatures lower polymer melt viscosity, allowing more material to be packed into mold to reduce shrinkage.

control shrinkage → increasing Amount of material that can be packed inside the MOLD → reducing shrinkage

by "playing" with MOLDING Temperature / Pressure of INJECTION

$T_M \uparrow$: easily to pack material (less viscouse) | $P_{INJ} \uparrow$ pack more material

↓ more material inside mold = less shrinkage

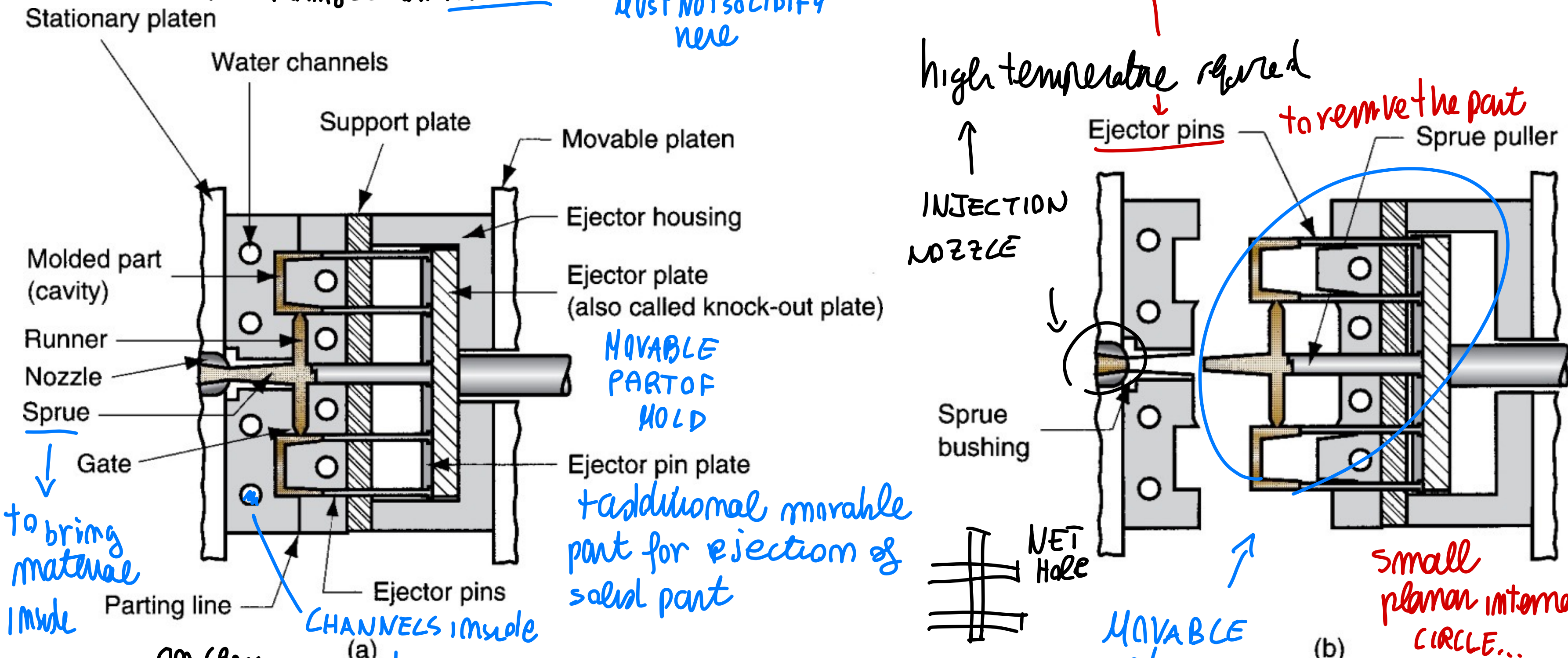
less SCHRINK if more material inside the MOLD

EXAMPLES of simple mold → in which we see fixed part inside the MOLD

The special tooling in injection molding – custom designed and fabricated for the part to be produced.

MOLD designed so that part go with movable half + ram for additional motion

FIXED PART related to injection NOZZLE → @ HIGH Temperature MUST NOT SOLIDIFY here



high temperature required

to remove the part

INJECTION NOZZLE

MOVABLE PART OF MOLD

+ additional movable part for ejection of solid part

to bring material inside

CHANNELS inside

NET HOLE

MOVABLE mold part

small planar internal CIRCLE... CYLINDRICAL PINS internally small

on CROSS SECTION

straight HOLES to create channels inside the mold → (to remove HEAT)

↳ different holes, NET between HOLES, circuit inside

THE MOLD

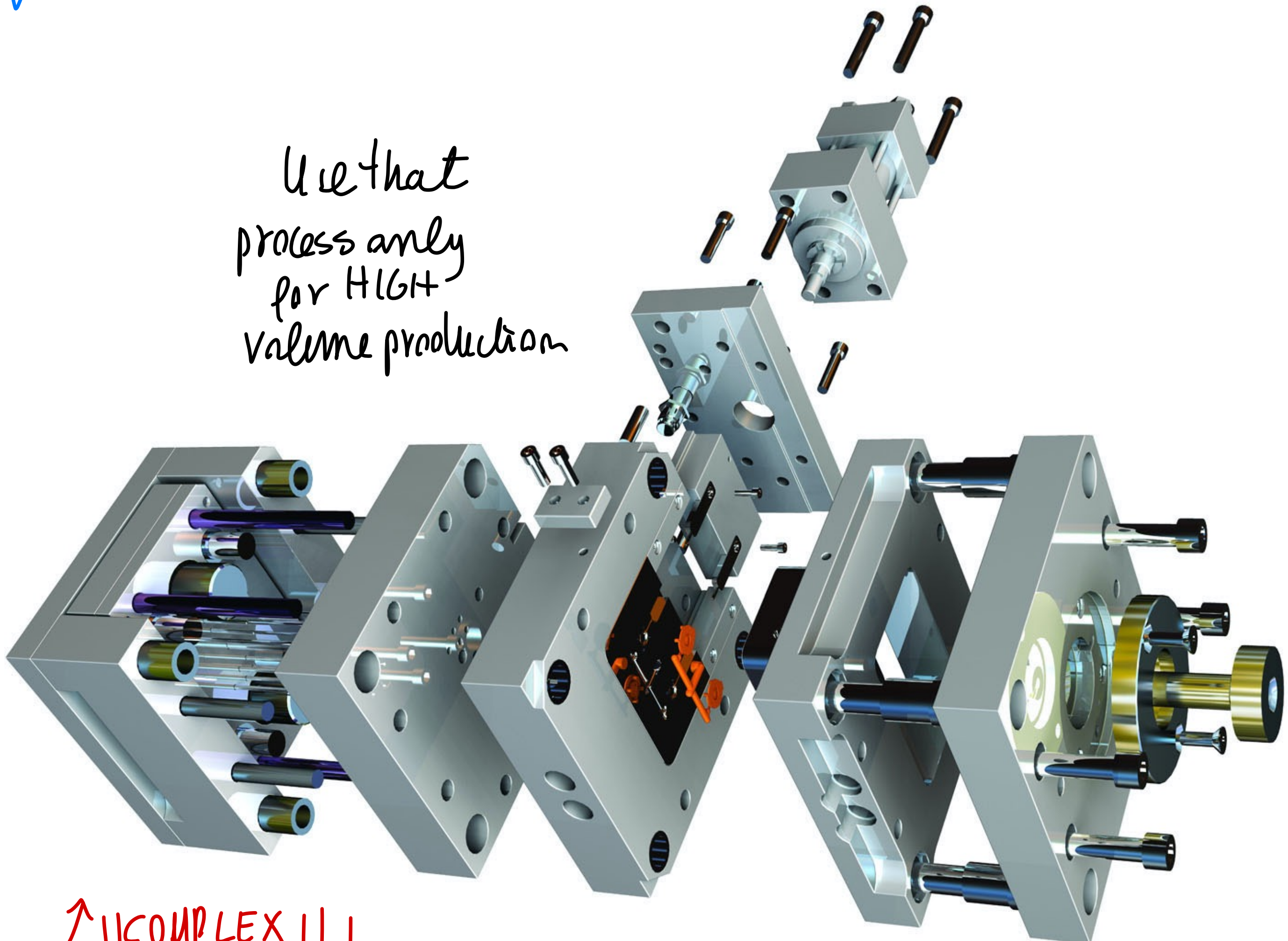
Net of hole → important to remove heat if (thermoplastic)

if (thermoset) → for HEATING material

(real MOLD design)

COMPLEX CASTING object

Use that process only for HIGH volume production



simple object

||COMPLEX MOLD|| !

ADDITIONAL POSSIBLE element to INJ

MULTICOMPONENT INJECTION MOLDING

additional element

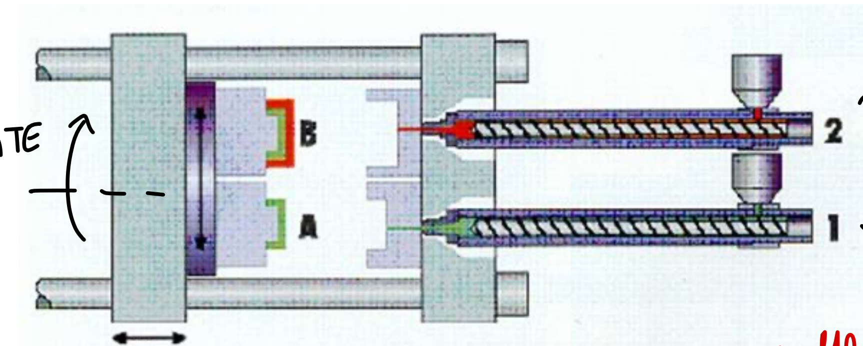
Unique polymeric PART

Co-Injection or Sandwich Molding

two injection units with two MOLDS and different materials

we can obtain additional cavity

ROTATE



2 INJECTION unit with different material

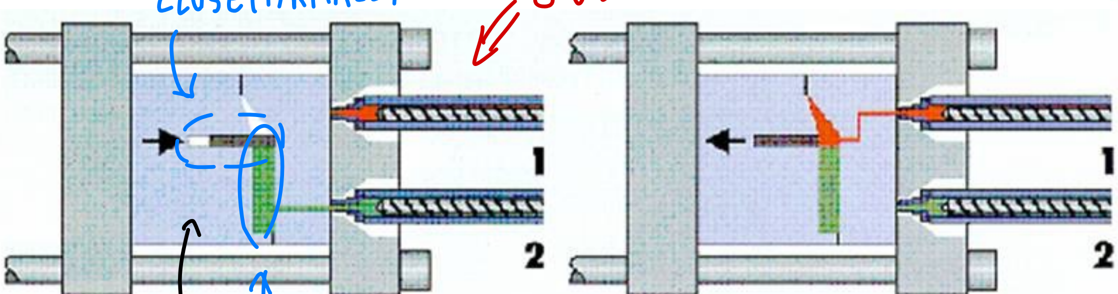
closing in CASE A we fill cavity with one polymer, then rotate the cavity we position B in A, close B and insert additional material

MULTI-MATERIAL OBJECT

LIKE to create different color/materials

CLOSE PARTIALLY

JUST ONE MOLD



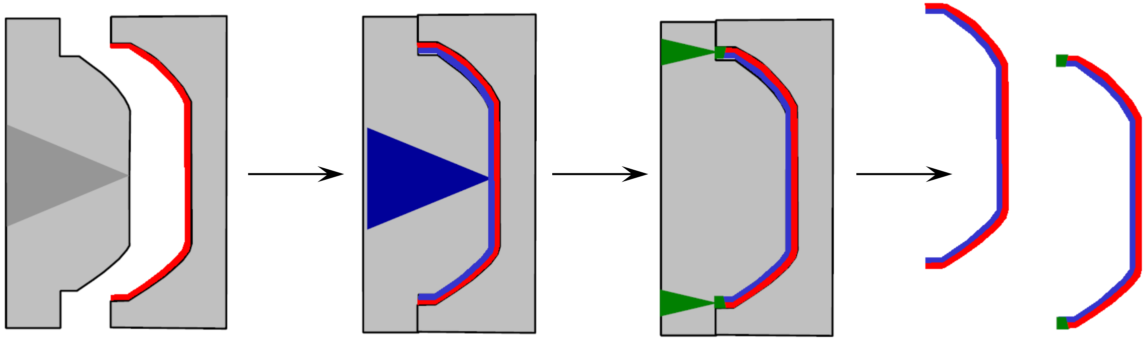
REMOVE

moovable elements, close mold partially

fill partially the mold, then additional space for second material

INJECTION MOLDING AND ASSEMBLING

Overmolding → different INJECTION units to fill the different parts



let additional material go over previous material

Cause different material on different portions

different portions the CAVITY filled with different material

so different INJECTION UNIT

INJECTION MOLDING OF THERMOSETS

let material be CURED inside the mold

Equipment and operating procedure must be modified to avoid premature cross-linking of TS polymer:

- Reciprocating-screw injection unit with shorter barrel length;
- Temperatures in barrel are relatively low.

Melt is injected into a heated mold, where cross-linking occurs to cure the plastic:

- Curing is the most time-consuming step in cycle;
- Mold is then opened, and part is removed.

on Thermosets → deal with pre-polymer

and let material CURE INSIDE MOLD

ROTATIONAL MOLDING

USE ROTATION to CREATE hollow closed part

USING a close MOLD but using ROTATION for HOLLOW CLOSED PART

Process uses gravity inside a rotating mold to produce a hollow part shape:

- Alternative to blow molding for larger parts with more complex external geometries
  - But lower production rates than blow molding
- Mostly for thermoplastic polymers but thermoset and elastomer applications are not uncommon
- Products: boat hulls, sandboxes, buoys, garbage cans, large containers and storage tanks

Rotational Molding

omogeneus layer on mold surface
(→ then on COOLING STATION (for thermoplastic)
(3)
COOL DOWN the mold STILL ROTATING
SOLIDIFY obtain LAYER

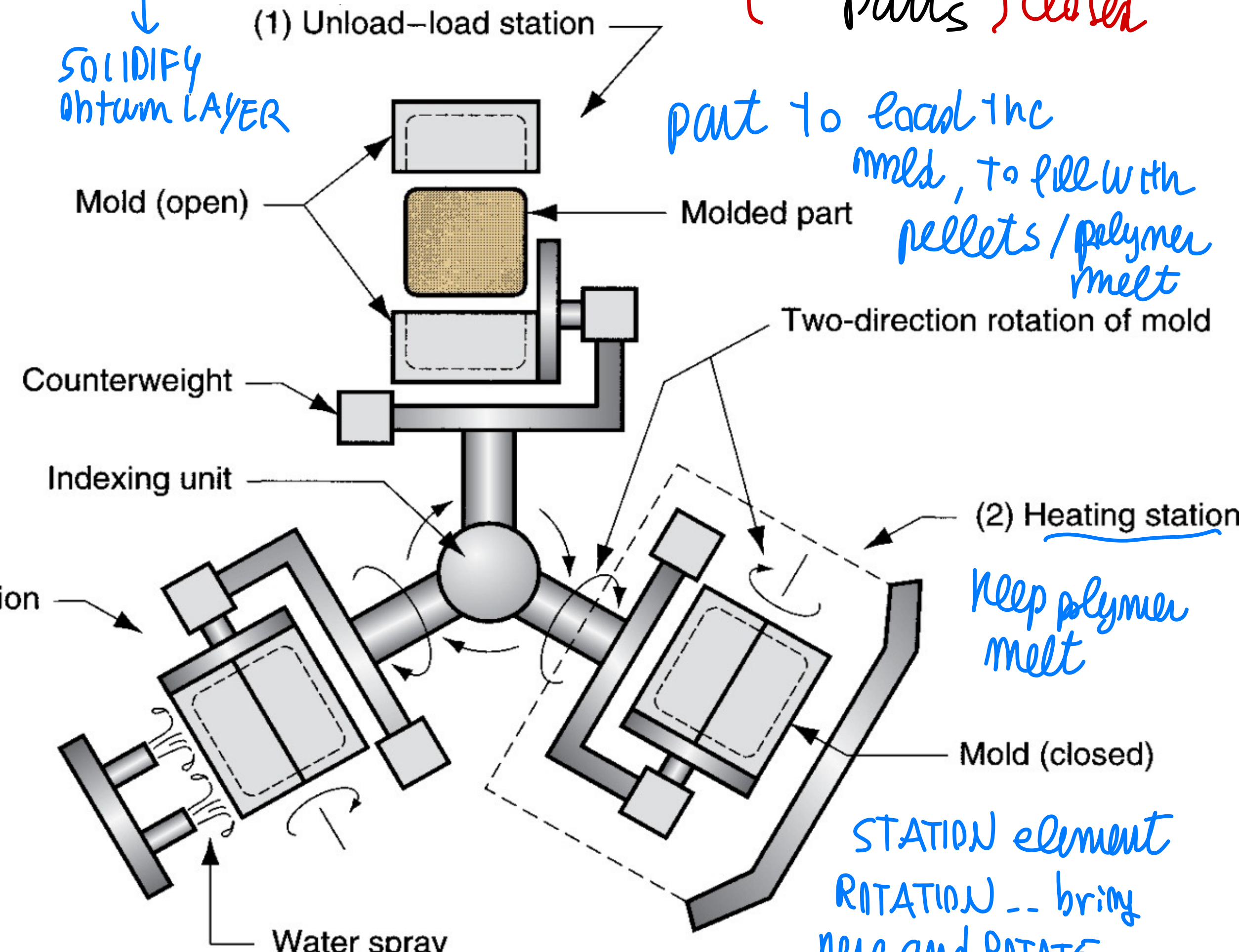
{3D Hollow parts} closed

Performed on three-station indexing table:

- (1) unload-load,
- (2) heat and rotate mold,
- (3) cool the mold.

different Heating

the polymer melts + create a uniform material



part to load the mold, to fill with pellets / polymer melt

keep polymer melt

STATION element ROTATION .. bring here and ROTATE mold heating material

due to ROTATION you create a uniform layer of material on mold surface → CAVITY INSIDE 3D Hollow PART

EXTRUSION BLOW MOLDING

permanent molds to create Hollow objects

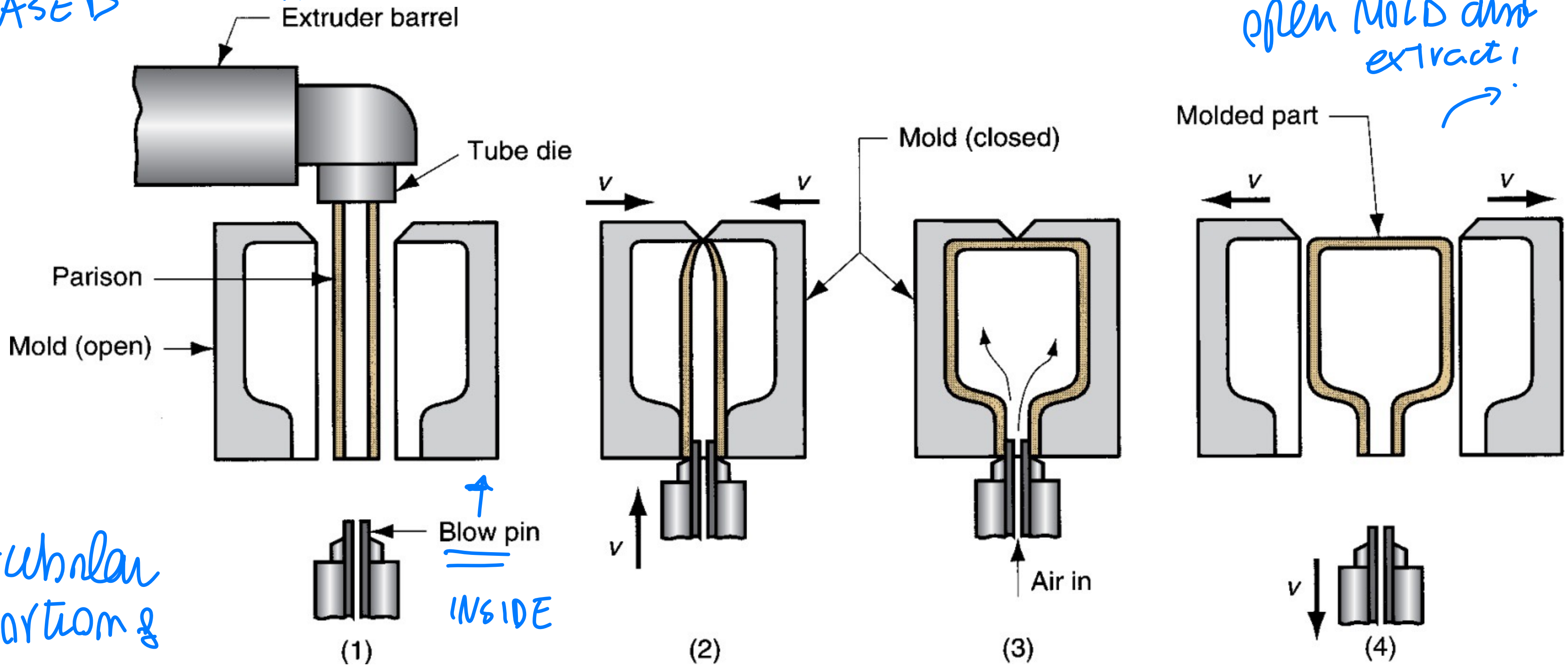
(1) Extrusion of parison; (2) parison is pinched at top and sealed at bottom around blow pin as the mold closes; (3) tube is inflated; and (4) mold is opened.

EXTRUSION based

EXTRUSION - BLOW MOLDING BASED

-> mold filled by extrusion

open MOLD and extract



tubular portion of material

Material Blown on the mold shape

material to feed mold comes from extrusion

from a TUBOLAR PARISON (portion of material)

TUBOLAR

close mold around tubolar material, enter the BLOW PIN to apply compresed air

and so material BLOW copying internal shape of mold

COSTANT thikness element

extracted @ full solidification

continuous extraction of parison

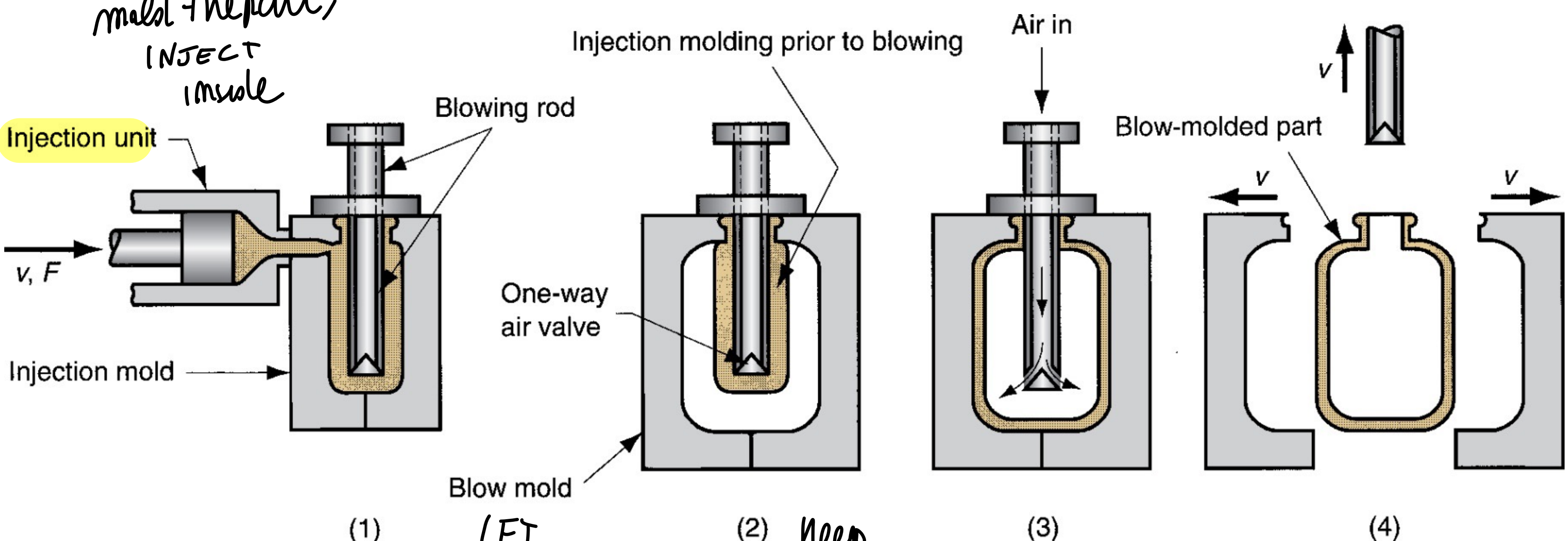
INJECTION BLOW MOLDING

simular system for Hollow compoment

(1) Parison is injected molded around blowing rod; (2) mold is opened, and parison is moved to blow mold; (3) soft polymer is inflated; and (4) mold is opened

similar OUTPUT, but by using INJECTION MOLDING

mold the part, INJECT inside



LET SOLIDIFY, then --

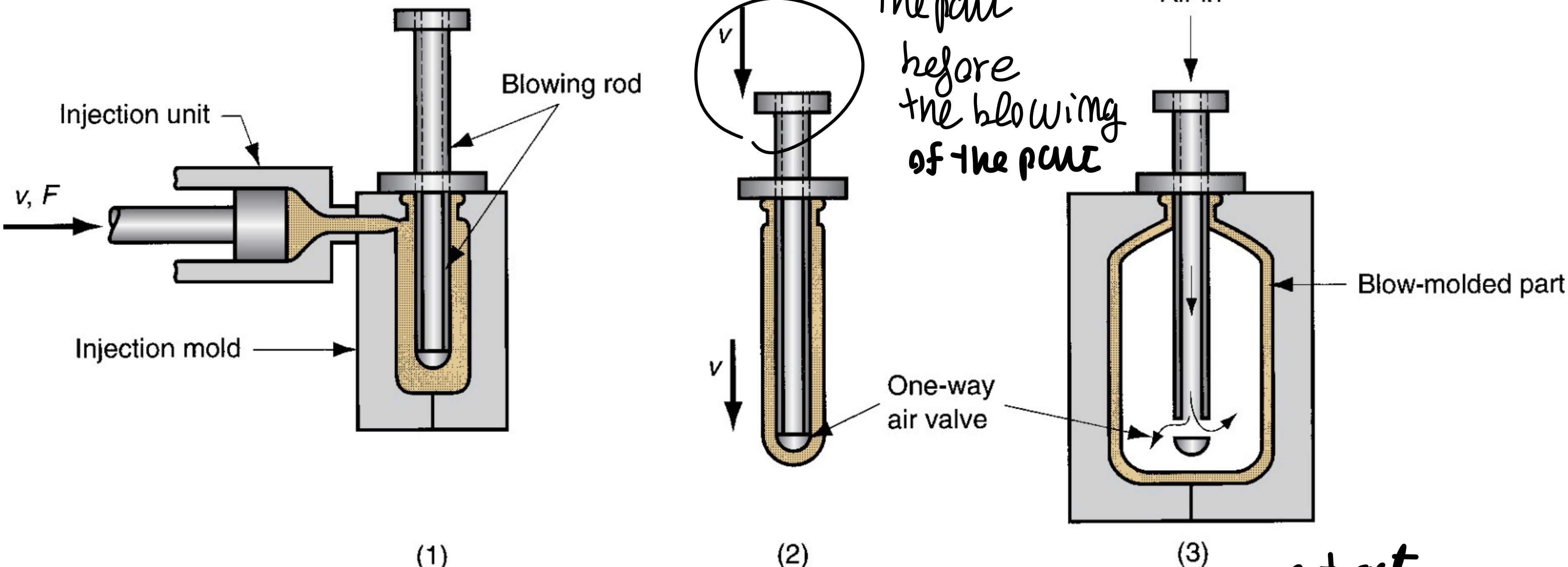
keep material enough deformable

↑ blow material through AIR to obtain desired shape

STRETCH BLOW MOLDING

simularly, here we "STRETCH"

(1) Injection molding of parison; (2) stretching; and (3) blowing



STRECH on the part before the blowing of the part

create a STRETCH on the part before blowing

{ common processes }

extract when full solidification...

MATERIALS AND PRODUCTS IN BLOW MOLDING

Blow molding is limited to ***thermoplastics***

Materials: high density polyethylene, polypropylene (PP), polyvinylchloride (PVC), and polyethylene terephthalate

Products: disposable containers for beverages and other liquid consumer goods, large shipping drums for liquids and powders, large storage tanks, gasoline tanks, toys, and hulls for sail boards and small boats.

EXAMPLES of previous processes...

applied on thermoplastics material only

ISSUES

Reaction Injection Molding

another kind of molding

Two highly reactive liquid ingredients are mixed and immediately injected into a mold cavity where chemical reactions leading to solidification occur.

we still have INJECTION but of different chemical elements that when in contact create chemical reaction inside mold

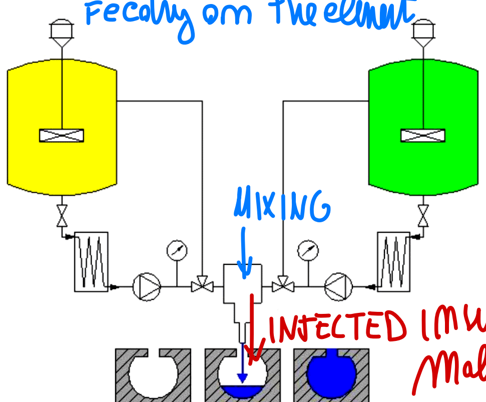
- RIM was developed with polyurethane to produce large automotive bumpers and fenders
  - RIM polyurethane parts have a foam internal structure surrounded by a dense outer skin

(CREATE FOAM)

- Other materials: urea-formaldehyde and epoxies

MIXING of chemical elements -> creating a chemical reaction... material with different density

Feeding on the element



MIXING

INJECTED inside mold

CREATE a chemical reaction

=

to polyurethane parts -> low/high density material

very different element

Polymer Foam

MIXING WITH AIR the Reaction polymers → you create FOAM/ SPONGES

A polymer-and-gas mixture that gives the material a porous or cellular structure:

- Most common polymer foams: polystyrene (Styrofoam, a trademark) and polyurethane
- Other polymers: natural rubber ("foamed rubber") and polyvinylchloride (PVC)

Properties of a Foamed Polymer

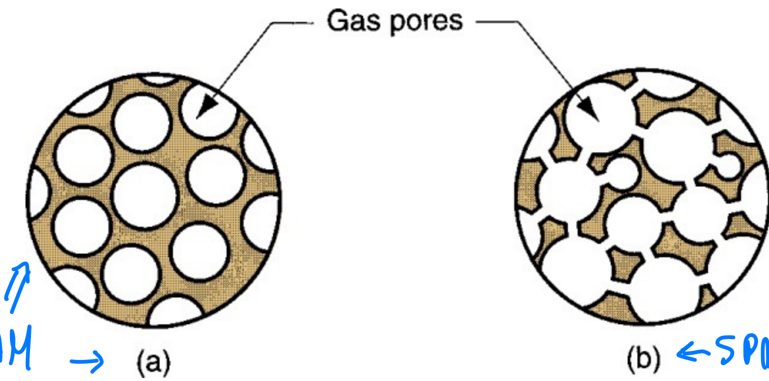
- Low density
- High strength per unit weight
- Good thermal insulation
- Good energy absorbing qualities

POLYMER FOAM

→ When air trapped inside the element (closed cells of AIR inside the object)

Two polymer foam structures: (a) closed cell and (b) open cell

{IF HIGH density: rigid material! NOT Foam/sponges}



FOAM → (a)

SPONGES if open cells

(b) ← SPONGES → high flexibility

- Elastomeric - matrix polymer is a rubber, capable of large elastic deformation
- Flexible - matrix is a highly plasticized polymer such as soft PVC
- Rigid - polymer is a stiff thermoplastic such as polystyrene or a thermoset such as a phenolic

- Applications: heat insulating structural materials, cores for structural panels, packaging materials, cushion materials for furniture and bedding, padding for automobile dashboards, and products requiring buoyancy

CASTING

you can apply CASTING → create polymer mold just by pouring of material inside the mold

Pouring liquid resin into a mold, using gravity to fill cavity, where polymer hardens

both on THERMOPLASTICS/SETS

- Both thermoplastics and thermosets are cast
  - Thermoplastics: acrylics, polystyrene, polyamides (nylons) and PVC
  - Thermosets: polyurethane, unsaturated polyesters, phenolics, and epoxies ← reaction chemical (curing) INSIDE the mold, NOT before!
- Simple mold
- Suited to low quantities

THERMOFORMING

(only on thermoplastic material)

particular process that, like sheet metal forming

Flat thermoplastic sheet or film is heated and deformed into desired shape using a mold: ↓ FORMING it

- Heating usually accomplished by radiant electric heaters located on one or both sides of starting plastic sheet or film;
- Widely used in packaging of products and to fabricate large items such as bathtubs, contoured skylights, and internal door liners for refrigerators.

Shape it by heat until flexible enough and soft enough

then cool down ⇒

thermoplastic of small thickness, sheet → Forming that shit

Big object but thin ↳ to shape it we heat the material, to be enough flexible to be formed

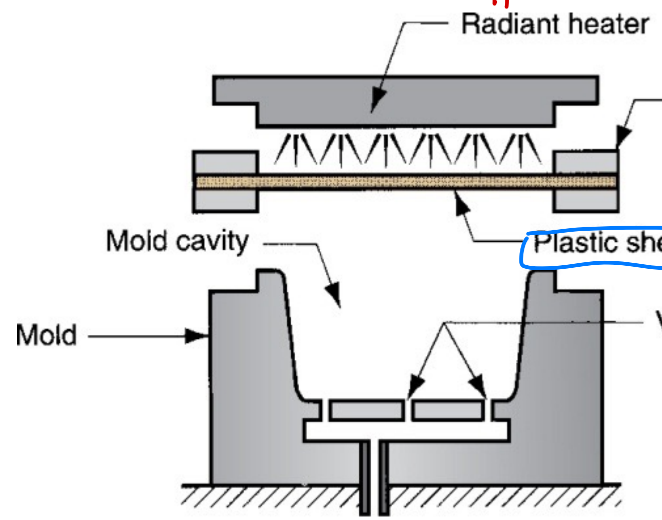
(Don't reach melt state) → soft enough and then cool down

VACUUM THERMOFORMING

to have soft material
shapable

(1) Flat plastic sheet is softened by heating

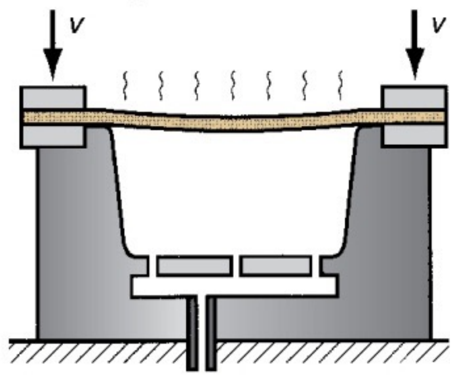
to heat the sheet



clamped on external boundaries

venting holes
to create vacuum inside cavity

(2) The softened sheet is placed over a concave mold cavity



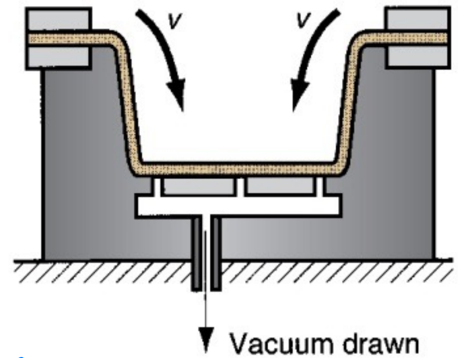
To deform material

VACUUM THERMOFORMING

(3) Vacuum draws the sheet into the cavity

↳ deform material and shape it like the mold

↳ material copy the shape of mold



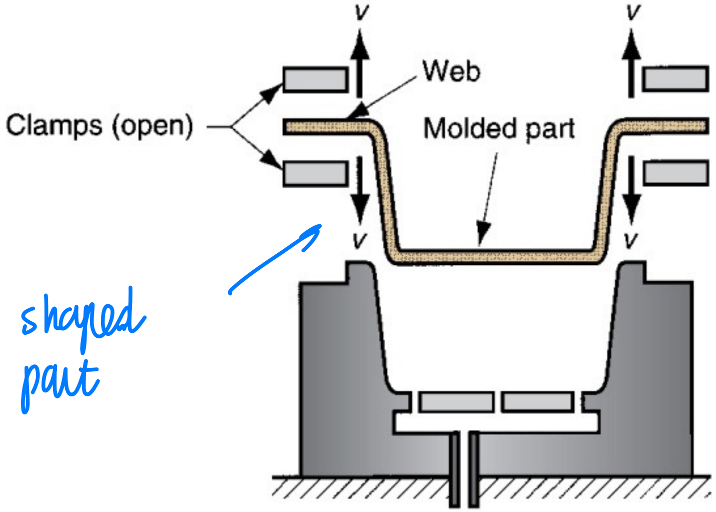
here just one component mold

controlling the cooling

(4) Plastic hardens on contact with the cold mold surface, and the part is removed and later trimmed from the web

wait for cooling down

shaped part

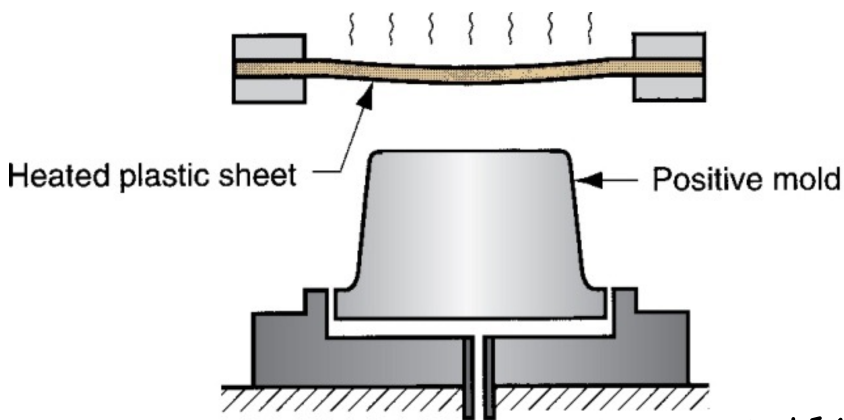


VACUUM THERMOFORMING USING A POSITIVE MOLD

We can copy the mold on a different way → stretch material around positive mold

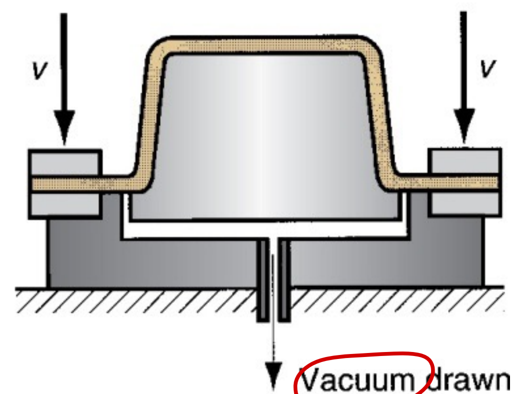
(1) Heated plastic sheet is positioned above the convex mold

(stretch around positive mold) → copy the geometry



VENTING HOLES to remove air and let material copy mold geometry

(2) Clamp is lowered into position, draping the sheet over the mold as a vacuum forces the sheet against the mold surface



between material and mold

Materials for Thermoforming

Only thermoplastics can be thermoformed

- Sheets of thermosetting or elastomeric polymers have already been cross-linked and cannot be softened by reheating

Common TP polymers: polystyrene, cellulose acetate, cellulose acetate butyrate, ABS, PVC, acrylic (polymethylmethacrylate - PMMA), polyethylene (PE), and polypropylene (PP).

common thermoplastic element for this process

different packaging ecc... different shapes can be obtained

MANY DIFFERENT OBJECT

as unique elements

THERMOFORMING

EXAMPLE...

Very complex shape tank can be created

also weld different material sheet, with separate mold

↓

WELD thanks to material high temperature

Uniform layers thikness!